

Climate Change and Pastoral Violence in Kenya

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Abstract

Due to the increasing effects of climate change on our atmosphere, climate and conflict research has become crucial in understanding and addressing the intricate relationships between environmental changes and social dynamics, providing insights for policy-making and sustainable development strategies. This thesis investigates the interplay between climate change, protected areas, and pastoral violence in Kenya, revealing how these elements contribute to the landscape of conflict. Through quantitative and spatial analysis of climate data, conflict incidents, and protected area boundaries, this study highlights the nuanced and multidimensional relationship between climate change and conflict. Going against previous research on climate and conflict in Africa, the results from this thesis show that high levels of rainfall are correlated with higher levels of conflict, for all types of conflicts and for pastoral conflicts, at the regional level and temporal level. These results suggest that livestock raiding plays a crucial role in pastoral violence across Kenya. Additionally, the research showed that the types of protected areas which had the highest concentrations of conflicts included community oriented protected areas and reserves. The thesis suggests that the dissemination of climate change information to pastoralists, inclusion of pastoral communities in the policy making process, and the creation of pastoral migratory corridors are a few of the many strategies which could help mitigate these conflicts in the future. This contribution to the discourse on climate change, conservation, and peace offers evidence-based strategies for sustainable development and conflict resolution in pastoral regions of Kenya.

Chapter 1: Introduction

“In the face of impending catastrophe, whose warning signs are already unbearably disastrous, weak action is unwise. No action is dangerous.”

William Ruto, President of Kenya¹

1.1 Pastoral Violence and Protected Areas in Kenya

Over the last three months, 30% of the violent attacks which have occurred in Kenya involved pastoralist militias and have resulted in at least 73 deaths. This accounted for a 77% increase in pastoral violence compared to previous years.² As pastoral violence has increased, Kenya has been simultaneously experiencing its worst drought on record over the past few dry seasons.³ With the majority of the population in Kenya’s drylands depending on pastoralism as their main source of income,⁴ climate change is only adding to the complex dynamics of pastoralism in this region.

In an interview with a pastoralist based in Northern Kenya, Mr. Metuy said: “The cows I have left are now like skeletons. They drag themselves through the field, waiting to die.”⁵

Pastoralism is a practice which involves herding livestock on free-range grassland to later be used for subsistence, sold in livestock markets, or traded for other useful items.⁶ Pastoralists are

¹ “COP27: The Top Quotes from Climate and World Leaders at the UN Summit,” World Economic Forum, November 11, 2022, <https://www.weforum.org/agenda/2022/11/cop27-quotes-climate-leaders/>.

² Communications ACLED, “Kenya Situation Update: March 2023 | Government Operation Against Pastoralist Militias in North Rift Region,” *ACLED* (blog), March 31, 2023, <https://acleddata.com/2023/03/31/kenya-situation-update-march-2023-government-operation-against-pastoralist-militias-in-north-rift-region/>.

³ “Kenya | Humanitarian Assistance,” U.S. Agency for International Development, March 6, 2024, <https://www.usaid.gov/humanitarian-assistance/kenya>.

⁴ “Climate Change Destroys the Livelihoods of Kenyan Pastoralists,” *Africa Renewal*, January 4, 2023, <https://www.un.org/africarenewal/magazine/january-2023/climate-change-destroys-livelihoods-kenyan-pastoralists>.

⁵ *Ibid.*

⁶ John G. Galaty, “Pastoralism in Anthropology,” in *International Encyclopedia of the Social & Behavioral Sciences (Second Edition)*, ed. James D. Wright (Oxford: Elsevier, 2015), 577–83, <https://doi.org/10.1016/B978-0-08-097086-8.12124-5>.

nomadic, meaning they are often traveling in search of fresh pasture and do not remain in the same place for too long.⁷ Pastoralism is a practice which has been ingrained in African culture for centuries, however it has recently become enmeshed in violent conflicts across Africa. This issue is especially prevalent in Kenya, due to its unstable political landscape and land ownership laws.⁸

Additionally, as the impacts of climate change have become more pronounced on the Kenyan environment, there has also been a recent resurgence of the creation of protected areas across Kenya, looking to create a harmonious balance between humans and wildlife.⁹ Recently, community conservancies have emerged as a method to promote wildlife conservation, mitigate land ownership conflicts, and support economic development of pastoral communities.¹⁰ Despite the efforts of conservancies to support pastoralists, these protected areas have been found to encroach on crucial lands which are used for pastoralism, which decreases available pasture and limits livestock mobility.¹¹ Additional attempts have been made to support pastoralists by integrating them into the tourism industry of conservancies, however their economic output was found to decrease compared to their original pastoral economic output, leaving them worse off.¹² These issues have given rise to a new set of violent conflicts involving protected areas and pastoralists.

⁷ “Pastoralism | Project Regeneration,” accessed April 1, 2024, <https://regeneration.org/nexus/pastoralism>.

⁸ Clemens Greiner, “Guns, Land, and Votes: Cattle Rustling and the Politics of Boundary (Re)Making in Northern Kenya,” *African Affairs* 112, no. 447 (April 1, 2013): 216–37, <https://doi.org/10.1093/afraf/adt003>.

⁹ Damiannah Kieti et al., “An African Dilemma: Pastoralists, Conservationists and Tourists – Reconciling Conflicting Issues in Kenya,” *Development Southern Africa* 37, no. 5 (September 2, 2020): 758–72, <https://doi.org/10.1080/0376835X.2020.1747988>.

¹⁰ Wu Liang et al., “The Development of Wildlife Community Conservancies in Kenya: A Preliminary Review,” *Journal of Resources and Ecology* 9, no. 3 (May 2018): 250–56, <https://doi.org/10.5814/j.issn.1674-764x.2018.03.004>.

¹¹ “Exploring Linkages between Protected-Area Access and Kenyan Pastoralist Food Security Using a New Agent-Based Model - Ecology & Society,” accessed April 1, 2024, <https://ecologyandsociety.org/vol29/iss1/art18/>.

¹² C. E. Bedelian, “Conservation, Tourism and Pastoral Livelihoods: Wildlife Conservancies in the Maasai Mara, Kenya,” *Doctoral Thesis, UCL (University College London)*. (Doctoral, UCL (University College London), 2014), <https://discovery.ucl.ac.uk/id/eprint/1434122/>.

In this thesis, I will explore the dynamics of pastoral violence in Kenya and how protected areas and climate change are facilitating this violence. Using spatial data on violent conflicts, protected area boundaries, and climate change variables, I will be conducting an analysis on how these three variables are interrelated. I will begin by exploring the correlation between climate change indicators and conflict at the temporal and spatial levels. Despite hypothesizing that greater effects of climate change will be associated with higher levels of conflicts, my results suggest that high rainfall and low drought levels are associated with more conflict, at the regional and annual level.

The next section analyzes how many conflicts are occurring within protected areas and which designation of protected areas have the highest rates of conflict. My results follow my hypothesis, and I find that protected areas focused on community conservation and reserves had higher levels of conflict compared to other types of protected areas. Finally, I analyze pastoral conflicts and how they are intertwined in the climate conflict dynamic. My results suggest that high levels of pastoral violence corresponds with higher rainfall, as hypothesized, and that pastoral violence has significantly increased in recent years. Additionally, I find that pastoral violence in protected areas occurs at higher rates in community based protected areas and reserves, which also follows my original hypothesis.

1.2 Motivation and Framework

In this section, I will explain my motivations for writing this thesis and for the framework which I will be employing in my research. One of my main motivations, climate change, is at the forefront of today's global issues, indicating a pressing need for change before irreversible

damage is done to the planet.¹³ Climate change has many direct effects, like longer droughts, increased temperature, and sea level rise.¹⁴ However, it also has many discrete effects as well, one being an increase in conflict around the world.¹⁵ Climate effects have disproportionately affected Africa, leaving it with extreme increases in temperature and droughts,¹⁶ which have also been associated with an increase in violent conflict across Africa.¹⁷

As a result of this, there has been a variety of research on climate change and conflict across Africa. Some research has found a causal link between climate change and conflict, while others find that conflict is a much more nuanced issue that can not simply be attributed to climate change.¹⁸ In this thesis, I will be examining the direct impact of climate change on conflict in Kenya in addition to other indirect pathways using variables like protected areas and pastoralism.

For the scope of my research, I chose to focus on Kenya because of a few different factors. Many conflict and climate studies have been conducted at the multi-national or the regional level, however there is a lack of climate and conflict studies which have been conducted at the national level.¹⁹ Examining the impact of climate change on conflict across multiple countries helps us understand large scale trends in conflict and can offer general recommendations which can be implemented in a variety of different environments. On the other hand, regional studies help us understand a more specific dynamic between climate change and

¹³“Climate Change | United Nations,” accessed April 1, 2024, <https://www.un.org/en/global-issues/climate-change>.

¹⁴ Ibid.

¹⁵ Ibid.

¹⁶“Africa Suffers Disproportionately from Climate Change - World | ReliefWeb,” September 4, 2023, <https://reliefweb.int/report/world/africa-suffers-disproportionately-climate-change>.

¹⁷ Eoin F McGuirk and Nathan Nunn, “Transhumant Pastoralism, Climate Change and Conflict in Africa,” December 14, 2021, https://scholar.harvard.edu/files/nunn/files/pastoralism_conflict_12_9.pdf.

¹⁸ Christian Almer, Jérémy Laurent-Lucchetti, and Manuel Oechslin, “Water Scarcity and Rioting: Disaggregated Evidence from Sub-Saharan Africa,” *Journal of Environmental Economics and Management*, Special issue on environmental economics in developing countries, 86 (November 1, 2017): 193–209, <https://doi.org/10.1016/j.jeem.2017.06.002>.

¹⁹ “Research Methods for Exploring the Links between Climate Change and Conflict - Ide - 2017 - WIREs Climate Change - Wiley Online Library,” accessed April 1, 2024, <https://wires.onlinelibrary.wiley.com/doi/full/10.1002/wcc.456>.

conflict, giving specific conflict mitigation solutions which can only be implemented in specific areas. This thesis blends both of these methods, focusing on the national level, and hopes to offer both large and small scale conflict mitigation strategies to reduce future conflicts.

Kenya's unique mix of political dynamics, pastoral traditions, and land ownership challenges presents a fascinating case for conflict analysis. Pastoralism is heavily integrated into the Kenyan economy and culture, putting it at the center of many recent violent conflicts.²⁰ Additionally, there has been a notable rise in livestock raiding incidents, where armed youth forcefully expropriate livestock herds from pastoralists, stripping them of their economic foundation.²¹ These acts of violence are further fueled by political instability, with the raiders frequently mobilized along ethnic and political lines.²² Furthermore, Kenya has a long and outstanding history of conflicts over land ownership, dating back to imperial rule of Kenya.²³ Presently, factors such as ethnic divisions, the scarcity of pasture, and the designation of protected areas are among the primary catalysts for land-related conflicts in the nation.

The final reason why I chose to focus on Kenya for my research is the recent increase in the country's community conservancies. During the summer of 2022, I spent two months at the Mpala Research Center, located in the Laikipia Province of central Kenya, an area with a high concentration of community conservancies. During this time, while researching water mitigation strategies in nearby communities, I was exposed to many of the issues which I will be focusing on in this paper, particularly the lack of agency given to the pastoral communities within

²⁰ Dickson Nyariki and Amwata Dorothy, "The Value of Pastoralism in Kenya: Application of Total Economic Value Approach," *Pastoralism* 9 (December 1, 2019), <https://doi.org/10.1186/s13570-019-0144-x>.

²¹ Janpeter Schilling, Francis EO Opiyo, and Jürgen Scheffran, "Raiding Pastoral Livelihoods: Motives and Effects of Violent Conflict in North-Western Kenya," *Pastoralism: Research, Policy and Practice* 2, no. 1 (December 3, 2012): 25, <https://doi.org/10.1186/2041-7136-2-25>.

²² Greiner, "Guns, Land, and Votes."

²³ Peter Veit, "History of Land Conflicts in Kenya," *Gates Open Res* 3, no. 982 (March 25, 2019): 982, <https://doi.org/10.21955/gatesopenres.1115885.1>.

protected areas. Community conservancies are a relatively new idea, aimed at providing communities with extra economic support and integration into the tourism industry, in return for taking land that would otherwise be used for farming or pastoralism.²⁴ Witnessing these tensions on the individual level, made me realize that these patterns were most likely occurring all across Kenya. Based on this experience, I wanted to focus on a general analysis of conflicts in Kenya, with a special focus on pastoral violence and protected areas. In the next section of the introduction, I will provide a roadmap for this thesis.

1.3 Roadmap

In the next chapter of this thesis, I will outline relevant literature and background information on the topic. I will begin by analyzing climate and conflict literature, then give background on protected areas and conflicts associated with them, and finally outline pastoral violence in Kenya and its potential causes. Chapter 3 will discuss the methodology of this thesis, focusing on the main variables, statistical tests, and specific data which will be used. In Chapter 4, I will begin my results section by analyzing the relationship between climate change and conflict. I will be using both temporal and spatial variables like annual rainfall, annual drought, regional rainfall, and regional phytomass to see how conflict intersects with an increase or decrease in these variables.

In Chapter 5, I will be using Quantum GIS(QGIS) modeling to count the number of conflicts which have occurred inside the boundaries of protected areas and the count of each designation of protected areas which had recorded conflicts. The final chapter of the results section will analyze the previous two variables with respect to pastoral violence. Using a subset

²⁴ “Green Grab by Bricolage - The Institutional Workings of Community Conservancies in Kenya on JSTOR,” accessed April 1, 2024, <https://www.jstor.org/stable/26500660?seq=11>.

of data which contains pastoralists as one of the main actors involved in conflicts, I will test for correlation between pastoral violence and climate change variables, like rainfall. Next, I will analyze the number of recorded pastoral conflicts within each type of protected area to see which type of protected area had the most recorded conflicts. Finally, in Chapters 7 and 8, I will discuss the results of my thesis, the implications of the results, and outline a set of policy recommendations geared towards reducing conflicts related to pastoralism and protected areas in Kenya.

Chapter 2: Literature Review

This chapter aims to address relevant literature and background necessary to understand the research being conducted in this thesis. To begin, I will outline climate change and conflict research, focusing on literature which aims to show a causal effect between climate change and conflict. Next, I will highlight a specific subset of climate and conflict research, resource scarcity, and explain how it is specifically applicable to Kenya. I will also discuss common methodology practices in this area of research to be used as a base for the methodology employed in this thesis.

In the next section, I will analyze protected areas as a variable involved in the climate conflict paradox. I will begin by introducing protected areas as a concept, then I will discuss the types of protected areas in Kenya, and finally I will examine research relating protected areas to conflict. The third section of this literature review will focus on how pastoral violence is involved in this dynamic and the many factors involved in pastoral conflicts in Kenya. Next, I will discuss recent empirical work pertaining to this thesis and the gaps in the literature which this thesis hopes to fill. Finally, I will conclude with stating my research question and hypotheses.

2.1 Climate Change and Conflict

As stated in the introduction, climate change remains one of the most pressing issues of our time, highlighting the need for adaptation and change in order to preserve our planet.²⁵

Despite its more direct effects, like erratic temperature and rising sea levels²⁶, climate change

²⁵ Lea Berrang-Ford, James D. Ford, and Jaclyn Paterson, “Are We Adapting to Climate Change?,” *Global Environmental Change* 21, no. 1 (February 1, 2011): 25–33, <https://doi.org/10.1016/j.gloenvcha.2010.09.012>.

²⁶ “Modern Global Climate Change | Science,” accessed April 1, 2024, <https://www.science.org/doi/full/10.1126/science.1090228>.

also has more discrete effects on human interactions. However, there is a lot of uncertainty surrounding the linkage between climate change and its social consequences, highlighting the need for research surrounding this topic.²⁷

Examining the relationship between climate change and conflict is extremely nuanced, highlighting the presence of many pathways and causal mechanisms linking the two.²⁸ There has been a wide variety of research linking climate change to conflicts, both interpersonal and intergroup, however these articles often also argue that conflict is contingent on a large number of other variables as well.²⁹ Some scholars try to explain direct pathways, like temperature and rainfall, and their impact on conflict and others look to explain more indirect pathways, like social or physiological ones. For example, Burke et al. (2015) takes a direct pathway approach and found that a 1 unit increase in temperature led to a 2.4% increase in interpersonal conflict and an 11.3% increase in intergroup conflict.³⁰ Rainfall also has exacerbating effects on conflict, with some scholars finding that years with high variability in rainfall had higher amounts of conflict, especially violent ones.³¹ Despite the data showing that rainfall and temperature are correlated with increasing levels of conflict, it is also important to take into account other, more nuanced causes of conflict associated with climate change like political and social mechanisms.³²

Exploring indirect pathways linking climate change and conflict can give us a more comprehensive understanding of this relationship. This relationship is also heavily context

²⁷ “Whither the Weather? Climate Change and Conflict,” accessed April 1, 2024, <https://doi.org/10.1177/0022343311431288>.

²⁸ “Climate Change, Conflict and Health - Devin C Bowles, Colin D Butler, Neil Morisetti, 2015,” accessed April 1, 2024, <https://journals.sagepub.com/doi/10.1177/0141076815603234>.

²⁹ Idean Salehyan, ““From Climate Change to Conflict? No Consensus Yet,” accessed April 1, 2024, <https://doi.org/10.1177/0022343308088812>.

³⁰ Marshall Burke, Solomon M. Hsiang, and Edward Miguel, “Climate and Conflict,” *Annual Review of Economics* 7, no. 1 (August 1, 2015): 577–617, <https://doi.org/10.1146/annurev-economics-080614-115430>.

³¹ “Climate Change, Rainfall, and Social Conflict in Africa - Cullen S Hendrix, Idean Salehyan, 2012,” accessed April 1, 2024, <https://journals.sagepub.com/doi/full/10.1177/0022343311426165>.

³² Idean Salehyan, “From Climate Change to Conflict? No Consensus Yet.”

dependent and often the setting can reduce or exacerbate conflicts. Some aspects that have been found to exacerbate the link between climate change and conflict are low human development, large populations, and political exclusion.³³ Additionally, economic output and migration can also be an indirect pathway to explain the increase in conflict.³⁴ As resources diminish, economic output decreases and leads individuals to move to a different location to maximize their profits.³⁵ Overall, there are many different indirect pathways which have been explored linking climate change to violent conflicts, and this thesis will attempt to highlight one as well, linking protected areas and pastoral violence to climate change and conflict. Before delving into specific pathways linking climate change to conflict, it is crucial to grasp the reasons behind Kenya's, and much of Africa's, status as a focal point for climate-related conflicts.

2.1.1 Climate Change and Conflict in Africa

The Intergovernmental Panel on Climate Change (IPCC) released its 6th synthesis report in 2023, which highlighted the areas where climate change vulnerability is highest:

“Vulnerability is higher in locations with poverty, governance challenges and limited access to basic services and resources, violent conflict and high levels of climate-sensitive livelihoods (e.g., smallholder farmers, pastoralists, fishing communities)”³⁶

The quote highlights that areas around the world with high levels of pastoralism, poverty, lack of resources, and violent conflicts are the most vulnerable to climate change. One area in which these issues are particularly prevalent is Africa.³⁷ During the past few decades, Africa has been

³³Tobias Ide et al., “Multi-Method Evidence for When and How Climate-Related Disasters Contribute to Armed Conflict Risk,” *Global Environmental Change* 62 (May 1, 2020): 102063, <https://doi.org/10.1016/j.gloenvcha.2020.102063>.

³⁴ Vally Koubi, “Climate Change and Conflict,” 2019.

³⁵ Ibid.

³⁶ “AR6 Synthesis Report: Climate Change 2023,” accessed April 1, 2024, <https://www.ipcc.ch/report/ar6/syr/>.

³⁷ FAO, *Pastoralism in Africa's Drylands: Reducing Risks, Addressing Vulnerability and Enhancing Resilience* (Rome, Italy: FAO, 2018), <https://www.fao.org/documents/card/en?details=ca1312en>.

disproportionately hit with detrimental climate shocks, including droughts, increased temperatures and tropical cyclones.³⁸

Despite having little to no carbon emissions, Africa is being impacted the most by climate change. These climate shocks, like increased temperatures and droughts, have negatively impacted economic growth, food security, and biodiversity, leaving Africa with a multitude of problems they didn't contribute to.³⁹ Additionally, many individuals in Africa are employed in sectors which are disproportionately affected by climate change, like agriculture and pastoralism.⁴⁰ There are over 268 million people in Africa who depend on pastoralism as their main source of livelihood,⁴¹ leaving many people employed by the pastoralist industry without proper access to water or fresh pasture, which in turn decreases their economic output.

Africa has also been affected by severe climate variation as a result of climate change. This has been represented by greater variation in rainfall, temperatures, and droughts which has had negative impacts on many pastoralist communities in Africa. Using satellite data and topographic fields, Funk et. al (2012) found that temperature and rainfall has varied more in recent years than ever before.⁴² In addition to the significant effects of climate change on its natural resources, Africa has also been affected by high levels of conflict, both inter- and intra-state. Research has found that drought and conflict are heavily correlated in most countries around the world, and especially in Africa. Many of these conflicts are attributed to instability within states and are driven by ethnic and political tensions.⁴³

³⁸“Africa Suffers Disproportionately from Climate Change - World | ReliefWeb.”

³⁹ Ibid.

⁴⁰ Ibid.

⁴¹ “Pastoral Ways - Africa Climate Mobility Initiative - Climate-Forced Migration,” Africa Climate Mobility Initiative, accessed April 1, 2024, <https://africa.climate-mobility.org/stories/>.

⁴²Brian D. Wardlow, Martha C. Anderson, and James P. Verdin, eds., “Mapping Recent Decadal Climate Variations in Precipitation and Temperature across Eastern Africa and the Sahel,” in *Remote Sensing of Drought*, 0 ed. (CRC Press, 2012), 354–81, <https://doi.org/10.1201/b11863-21>.

⁴³Norman Mlambo, *Violent Conflicts, Fragile Peace* (Adonis & Abbey Publishers Ltd, 2008).

However, there has also been research linking climate variation and sometimes even higher rainfall with increased conflicts. Hendrix et. al (2012) used a large-N statistical analysis of over 6,000 social conflicts and found a variety of different trends relating climate change to conflict in Africa. One of their major findings was that rainfall variability is correlated with political conflict. They also found that rainfall correlates with civil war and insurgency, and years with higher rainfall had more violent conflicts.⁴⁴ Overall, the results from most climate change and conflict studies in Africa indicate that higher effects of climate change are correlated with higher conflict levels. In the next section, I will be exploring a potential reasoning for this correlation, which is resource scarcity.

2.1.2 Resource Scarcity

One of the more direct pathways associating climate change with conflict is resource scarcity.⁴⁵ Resource scarcity occurs when the demand of a certain resource is high, but the supply of the resource is low.⁴⁶ Scholars argue that climate change significantly reduces the supply of natural resources, especially in vulnerable countries, which leaves many individuals without access to natural resources to sustain their livelihoods, despite their high demand. This, in turn, has both direct and indirect effects on conflict.⁴⁷ Maxwell and Reuveny (2000) created a model which explores the dynamics between resource scarcity and conflict. They begin by explaining how per capita availability of resources can lead to an increase in conflict. From there, conflicts can further deplete resources which causes a vicious cycle which hinders economic growth and

⁴⁴ “Climate Change, Rainfall, and Social Conflict in Africa - Cullen S Hendrix, Idean Salehyan, 2012.”

⁴⁵ Koubi, “Climate Change and Conflict.”

⁴⁶ “Scarcity,” accessed April 3, 2024, <https://education.nationalgeographic.org/resource/scarcity>.

⁴⁷ Jon Barnett and W. Neil Adger, “Climate Change, Human Security and Violent Conflict,” *Political Geography*, Climate Change and Conflict, 26, no. 6 (August 1, 2007): 639–55, <https://doi.org/10.1016/j.polgeo.2007.03.003>.

peacebuilding efforts.⁴⁸ This model is particularly prevalent when analyzing conflicts in Africa, because they are currently facing the highest levels of resource scarcity and lack the economic and political stability needed to limit conflicts and encourage peacebuilding.⁴⁹

One of the main resources whose availability hinges on climate change and incites large amounts of conflict is water. Recently, Africa was hit with its worst drought in over 40 years, leaving around 20 million people with food security issues.⁵⁰ The recent increase in droughts across Africa has prompted a variety of research on how water scarcity has led to an increase in conflicts. As expected, water scarcity has been found to have a direct impact on conflict. Almer et al. (2017) found that a one unit decrease in water availability in Sub-Saharan Africa led to an 8.3% increase in conflicts, indicating a correlation between water scarcity and conflicts. In this scenario, limited access to water increased competition which in turn increased conflicts.⁵¹ However, this relationship between water scarcity and conflict can have many other confounding factors as well, highlighting the importance of analyzing this relationship and the other factors involved in more detail.

In addition to limited water, droughts have also caused a decline in fresh pasture, which is negatively affecting those involved in the pastoral industry. As many pastoralists are left without access to fresh pasture for their livestock, they are forced to graze on other lands, often ones that they are not allowed on, which increases conflicts across Africa.⁵² Moreover, pastoral conflicts

⁴⁸ John W. Maxwell and Rafael Reuveny, "Resource Scarcity and Conflict in Developing Countries," accessed April 1, 2024, <https://doi.org/10.1177/0022343300037003002>.

⁴⁹ "Resource Scarcity: Which Countries Are Most Vulnerable?," Vision of Humanity, September 26, 2020, <https://www.visionofhumanity.org/resource-scarcity-hotspots-which-countries-are-the-most-vulnerable/>.

⁵⁰ Laura Paddison, "Catastrophic Drought That's Pushed Millions into Crisis Made 100 Times More Likely by Climate Change, Analysis Finds," CNN, April 27, 2023, <https://www.cnn.com/2023/04/27/africa/drought-horn-of-africa-climate-change-intl/index.html>.

⁵¹ Almer, Laurent-Lucchetti, and Oechslin, "Water Scarcity and Rioting."

⁵² Karim Hussein, James Sumberg, and David Seddon, "Increasing Violent Conflict between Herders and Farmers in Africa: Claims and Evidence," *Development Policy Review* 17, no. 4 (December 1999): 397–418, <https://doi.org/10.1111/1467-7679.00094>.

are becoming more present in Kenya, because of its high dependence on pastoralism in many rural communities.⁵³ Therefore, one pathway of the climate conflict dynamic that I will be researching in this thesis is pastoral violence. This issue proves to be a nuanced and multidimensional one, however gaining a better understanding of pastoral violence in Kenya could offer potential solutions to help mitigate conflicts involving pastoralists across all of Africa.

2.1.3 Methodology used in Conflict and Climate Research

In this section, I will outline common methodology used in conflict and climate research which I will be utilizing in this thesis. The majority of research which has attempted to find a causal link between climate change and conflict has used large-N statistical analysis as their methodology, which uses regression models to predict correlation between two variables. For example, most of these studies analyze the effect of a one unit increase of a climate change variable, like annual rainfall, on the amount of recorded conflicts in a given area.⁵⁴

An innovative addition to the large-N statistical analysis methodology is a two step analysis, which zooms in on specific pathways linking one or two different mechanisms to climate change and conflict.⁵⁵ For example, Buhaug et. al (2015) explores how climate variation is correlated with conflicts through domestic food production.⁵⁶ Another addition to the large-N analysis is using an interaction term to analyze the relationship between climate change and conflict. The interaction effect can help understand how the strength of the interaction between

⁵³ Nyariki and Dorothy, “The Value of Pastoralism in Kenya.”

⁵⁴ Tobias Ide, “Research Methods for Exploring the Links between Climate Change and Conflict - Ide - 2017 - WIREs Climate Change - Wiley Online Library.”

⁵⁵ Tobias Ide, “Research Methods for Exploring the Links between Climate Change and Conflict - Ide - 2017 - WIREs Climate Change - Wiley Online Library.”

⁵⁶ Todd Graham Smith, “Feeding Unrest: Disentangling the causal relationship between food price shocks and sociopolitical conflict in urban Africa,” accessed April 1, 2024, <https://doi.org/10.1177/0022343314543722>.

climate change and conflict differs during varying levels of a third variable.⁵⁷ For example, in areas with less development, Salehyan et. al (2014) found that the correlation between climate change and political violence is stronger.⁵⁸ In this example, development levels represent the interaction term, so the authors analyzed the correlation strength between climate change and political violence depending on the level of development.

Large-N studies usually vary in terms of scope, however many of those which examine climate and conflict trends across multiple countries find that there are other major drivers of conflicts, rather than just climate change.⁵⁹ These studies have high external validity, which means inferences can be widely applied, however they lack internal validity, meaning that causation is difficult to prove.⁶⁰ For large-N studies which occur at the subnational level, the results are less likely to show large scale, more robust trends in conflict, and will more likely show small scale trends in specific locations.⁶¹ For example, in a section of Northern Kenya, a spatial conflict analysis found that higher rainfall levels correspond to higher conflicts. This was attributed to cattle raiding, which favors areas of higher rainfall, due to the strategic placement of violence and the opportunistic values of cattle raiders.⁶² This small scale spatial analysis found an opposite trend to that of most climate conflict studies and it offered an explanation for what is driving conflicts in a select location.

⁵⁷ Tobias Ide, “Research Methods for Exploring the Links between Climate Change and Conflict - Ide - 2017 - WIREs Climate Change - Wiley Online Library.”

⁵⁸ Idean Salehyan and Cullen S. Hendrix, “Climate Shocks and Political Violence,” *Global Environmental Change* 28 (September 1, 2014): 239–50, <https://doi.org/10.1016/j.gloenvcha.2014.07.007>.

⁵⁹ Ibid; Tobias Ide et al., “On Exposure, Vulnerability and Violence: Spatial Distribution of Risk Factors for Climate Change and Violent Conflict across Kenya and Uganda,” *Political Geography*, Special Issue: Climate Change and Conflict, 43 (November 1, 2014): 68–81, <https://doi.org/10.1016/j.polgeo.2014.10.007>.

⁶⁰ Tobias Ide, “Research Methods for Exploring the Links between Climate Change and Conflict - Ide - 2017 - WIREs Climate Change - Wiley Online Library.”

⁶¹ Ibid.

⁶² Adrien Detges, “Close-up on Renewable Resources and Armed Conflict: The Spatial Logic of Pastoralist Violence in Northern Kenya,” *Political Geography* 42 (September 1, 2014): 57–65, <https://doi.org/10.1016/j.polgeo.2014.06.003>.

Combining both of these methods, some studies like that of Detges et. al (2016) analyze climate change and conflict at a large scale and then analyze the same relationship at a smaller scale using a specific subset of conflicts. For this study, the authors explored how rainfall affected armed violence depending on infrastructure development. They found that in areas with poorly developed road infrastructure the correlation between rainfall and armed violence was stronger than it was in the larger subset of conflicts that they analyzed.⁶³ This analysis was useful because it allowed the authors to analyze both large and small scale trends in conflicts.

This thesis incorporates the various methodologies outlined previously to thoroughly explore the intricate relationship between climate change and conflict in Kenya. My approach includes employing a large-N statistical analysis at both the national level and within a targeted subset of pastoral conflicts. Furthermore, I will employ a two-step model to examine the dynamics between protected areas and pastoral violence, integrating political violence as an interaction term to add depth to our understanding of these complex interactions. In the next section, I will introduce protected areas, as it is one of my variables in the two step model which I will be analyzing.

2.2 Protected Areas

Due to recent increases in conservation efforts across the world, protected areas have been created as a vehicle to limit human interaction and allow biodiversity to prosper in an uninterrupted manner.⁶⁴ However, protected areas have also become integrated into the human

⁶³ Adrien Detges, “Local Conditions of Drought-Related Violence in Sub-Saharan Africa: The Role of Road and Water Infrastructures,” *Journal of Peace Research* 53, no. 5 (September 1, 2016): 696–710, <https://doi.org/10.1177/0022343316651922>.

⁶⁴ “The Performance of Protected-Area Expansions in Representing Tropical Andean Species: Past Trends and Climate Change Prospects | Scientific Reports,” accessed April 1, 2024, <https://www.nature.com/articles/s41598-022-27365-7>.

economy through tourism. Nature-based tourism, including visits to these areas, has emerged as a significant economic activity, offering both socio-economic benefits and challenges to local communities. Protected areas can offer many benefits to individuals including employment and income, however they also have many detrimental effects on communities and groups of people like acculturation, price inflation, and conflict.⁶⁵

The integration of community support within the management and conservation strategies of protected areas is essential for their long term sustainability and success, especially in countries vulnerable to civil unrest. In order to properly integrate communities into protected areas, it is important to understand their perceptions of the risks and benefits. Oftentimes, community members will support protected areas and conservation efforts as long as they don't interfere or encroach on their own land. Vodouhe et al. (2010) found that communities in Benin were mostly in support of protected areas, so long as they didn't limit the agricultural production of nearby communities.⁶⁶ Additionally, stable land ownership and the ability to continue to exercise their own form of governance are other variables which are essential aspects which are necessary to promote a harmonious relationship between protected areas and communities. For this thesis, I will be focusing on protected areas in Kenya, however it is important to keep in mind trends from protected areas and communities around the world when considering solutions.

⁶⁵ "Nature-Based Tourism in Protected Areas: A Systematic Review of Socio-Economic Benefits and Costs to Local People | Semantic Scholar," accessed April 1, 2024, <https://www.semanticscholar.org/paper/Nature-based-tourism-in-protected-areas%3A-a-review-Thapa-King/edbb2fd95cfd97cc42f0e8781f50e241ceb1dfe>.

⁶⁶ Fifanou G. Vodouhê et al., "Community Perception of Biodiversity Conservation within Protected Areas in Benin," *Forest Policy and Economics* 12, no. 7 (September 1, 2010): 505–12, <https://doi.org/10.1016/j.forpol.2010.06.008>.

2.2.1 Protected Areas and Conflict

Protected areas, while crucial for biodiversity conservation, often become epicenters of conflict between conservation objectives and local community interests. These conflicts manifest in various forms, ranging from disputes over land rights to disagreements on the use of natural resources. A deeper understanding of these conflicts, their roots, and potential resolution strategies is vital for the sustainable management of protected areas.

In India, the push for biodiversity conservation, particularly with initiatives focused on tiger protection, has led to heightened conflicts. The Wildlife Protection Act of 1972 and subsequent conservation policies have sometimes aggravated tensions between local communities and forest guards. Fanari (2022) discusses the political ecology of conservation conflicts in India, highlighting the socio-economic impacts of conservation policies, including dispossession, violence, and the rise of "green militarization." It also focuses on the tensions that arise between conservation initiatives and the fight over common pool resources, like land or water. This scenario underscores the need for a balanced approach that respects the rights and livelihoods of local communities while achieving conservation goals.⁶⁷

The complexity of conservation conflicts is further illustrated by Hodgson et al. (2022), who explore stakeholder responses to conflicts over grouse shooting and raptor conservation. Their study identifies three main narratives—cooperation, resistance, and despondence—demonstrating the varied ways individuals justify their responses to conflict. They explain that current protected area management techniques prompt communities to respond with resistance, rather than cooperation or despondence. This suggests that effective conflict management must go beyond reconciling disputes to address the deeper-rooted issues that fuel

⁶⁷ Eleonora Fanari, "Struggles for Just Conservation: An Analysis of India's Biodiversity Conservation Conflicts," *Journal of Political Ecology* 28, no. 1 (August 11, 2022), <https://doi.org/10.2458/jpe.5214>.

these disagreements.⁶⁸ By adopting these strategies, conservation efforts can move towards more sustainable and equitable outcomes that respect both biodiversity and the rights and livelihoods of local populations.

2.2.2 Protected Areas in Kenya

There are many different forms of protected areas in Kenya, and the majority of them are managed by Kenya Wildlife Services (KWS). KWS categorizes a protected area as “landscapes /seascapes that have been surveyed, demarcated and gazetted either as National Parks and/or National Reserves.”⁶⁹ Due to Kenya’s biodiversity, these areas can be found across many different biomes and ecosystems. National parks don’t allow any sort of human interactions other than tourism and research, while national reserves allow limited amounts of human interaction, like recreational fishing.⁷⁰ It is important to note that most protected areas in Kenya are not fully fenced, in order to allow wildlife to roam freely, however this also allows humans to utilize and take protected area resources due to the lack of physical barriers.⁷¹

Across Kenya, there are 23 national parks, all of which are managed by KWS. Some of the main challenges which KWS faces when managing these national parks are declines in revenue, declines in tourism, poaching, and infrastructure loss.⁷² In addition to these challenges, national parks in Kenya also have to account for conflicts between humans and wildlife and conflicts between the various types of protected areas and surrounding communities. Western et al. (2015) explores the relationships between humans, communities, national park management,

⁶⁸ “Fight or Flight? Understanding Different Stakeholder Responses to Conservation Conflicts,” accessed April 1, 2024, <https://www.tandfonline.com/doi/full/10.1080/08941920.2022.2048933>.

⁶⁹ “Overview | Kenya Wildlife Service,” accessed April 1, 2024, <https://www.kws.go.ke/content/overview-0>.

⁷⁰ Ibid.

⁷¹ Ibid.

⁷² Yvonne Murichu and Lydia Gachengo, “Total Quality Management Practices and Performance of Kenya Wildlife Service in Nairobi County, Kenya,” *International Journal of Business Management, Entrepreneurship and Innovation* 5, no. 3 (October 8, 2023): 36–54, <https://doi.org/10.35942/k8rqt477>.

and wildlife in Kenya. Recently, there has been an increase in national park coverage on human modified land and land with higher levels of human interaction. Thus, it is important to ensure that landowners are accommodating to wildlife and that communities are involved in national park management and conservation efforts. This solution targets a stable coexistence between national parks, communities, and wildlife.⁷³

Another type of protected area which can be found all across Kenya's landscape are reserves. The majority of reserves in Kenya, amounting to 243 total, are forest reserves which can be located on trust land and unalienated government land and are demarcated and gazetted.⁷⁴ The forest reserves which are on trust land are managed by Kenya Forest Service(KFS) and the forest reserves which are on government land are managed by local authorities. There is limited research on forest reserves across Kenya, however their main goal is to preserve the rapidly declining tree coverage across the country.⁷⁵

Forest reserves and national reserves differ from national parks because they allow for recreational use, like fishing or charcoal production. However, recreational use can also cause conflicts because of the limited amount of resources in each area and the unclear land use policies surrounding land ownership and possession of resources within the reserves.⁷⁶ In addition, forest reserves also possess high levels of vegetation compared to other areas in Kenya,

⁷³ David Western, John Waithaka, and John Kamanga, "Finding Space for Wildlife beyond National Parks and Reducing Conflict through Community Based Conservation: The Kenya Experience," *PARKS* 21, no. 1 (March 27, 2015): 51–62, <https://doi.org/10.2305/IUCN.CH.2014.PARKS-21-1DW.en>.

⁷⁴ "FOREST COVER AND FOREST RESERVES IN KENYA: POLICY AND PRACTICE," accessed April 1, 2024, <https://portals.iucn.org/library/sites/library/files/documents/2000-019-05.pdf>.

⁷⁵ "Full Article: What Rights and Benefits? The Implementation of Participatory Forest Management in Kenya: The Case of Eastern Mau Forest Reserve," accessed April 1, 2024, <https://www.tandfonline.com/doi/full/10.1080/10549811.2017.1289105>.

⁷⁶ Godwin Leslie Muhati, Daniel Olago, and Lydia Olaka, "Land Use and Land Cover Changes in a Sub-Humid Montane Forest in an Arid Setting: A Case Study of the Marsabit Forest Reserve in Northern Kenya," *Global Ecology and Conservation* 16 (October 1, 2018): e00512, <https://doi.org/10.1016/j.gecco.2018.e00512>.

so pastoralists will often utilize the fresh pasture in the forest reserves and illegally graze on the land, creating further conflicts.⁷⁷

Community conservancies are a new addition to the protected area system in Kenya, which have been introduced with the intention of integrating community members into the tourism industry, reducing conflict, and increasing biodiversity.⁷⁸ Community conservancies in Kenya are managed by the Kenya Wildlife Conservancy Association (KWCA) and can be defined as “a recognized land use, making it an attractive land use option for communities and landowners as it offers improved land and resource rights and access to real incentives.”⁷⁹ One organization which has spearheaded the movement towards community centered conservation efforts is the Northern Rangelands Trust (NRT). They own and lead 33 conservancies across Kenya in an attempt to support indigenous communities.⁸⁰

Despite community conservancies being directed towards helping community members, they can also have negative effects on communities. Bersaglio and Cleaver (2018) explain how community conservancies create a contradictory environment because they attempt to be adaptive and progressive, however they are still constrained by the remnants of colonialism and land use issues. There are many historical remnants of European colonialism in Kenya which are still present today and heavily integrated into land use policies and rules which community conservancies abide by.⁸¹ In addition, community conservancies “reorganize the pastoralist commons” leading to many political and social consequences.⁸² Therefore, while community

⁷⁷ Ibid.

⁷⁸ Liang et al., “The Development of Wildlife Community Conservancies in Kenya.”

⁷⁹ “Conservancies in Kenya | KWCA,” August 23, 2016, <https://kwcakenya.com/conservancies/>.

⁸⁰ “Northern Rangelands Trust | KWCA,” September 1, 2017, <https://kwcakenya.com/regional-associations/northern-rangelands-trust/>.

⁸¹ “Green Grab by Bricolage - The Institutional Workings of Community Conservancies in Kenya on JSTOR.”

⁸² Annemiek Pas, Elizabeth E. Watson, and Bilal Butt, “Land Tenure Transformation: The Case of Community Conservancies in Northern Kenya,” *Political Geography* 106 (October 1, 2023): 102950, <https://doi.org/10.1016/j.polgeo.2023.102950>.

conservancies appear to be directed towards conservation and community centric ideals, they also perpetuate long lasting issues contributing to conflicts across the country.

Overall, protected areas have both negative and positive effects on communities in Kenya. When implemented properly, they can support communities and offer an alternative source of income. However, protected areas also have the potential to perpetuate tensions over politics and land ownership which have been manifesting for centuries and can further strengthen the divide between protected areas and communities. In the next section, I will discuss the final variable which I will be analyzing in this thesis, pastoral violence.

2.3 Pastoral Violence

Pastoralism, a livelihood widely practiced in Kenya, involves the nomadic practice of herding livestock in search of pasture.⁸³ Over 268 million people across the African continent depend on pastoralism as their livelihood, making it one that is heavily integrated into the African culture and economy.⁸⁴ The recent increase in pastoral violence can be attributed to a few causes: resource scarcity, political instability, land ownership, and protected areas.⁸⁵ This section will explore the various causes attributed to this recent increase in violence and investigate research that has been conducted on pastoral violence in Kenya.

2.3.1 Pastoral Violence and Climate Change

As drought and extreme weather events increase in Kenya, access to clean water and pasture is becoming more limited.⁸⁶ Resource scarcity is an issue which is inciting conflicts all

⁸³ “Pastoralism | Project Regeneration.”

⁸⁴ FAO, *Pastoralism in Africa's Drylands*.

⁸⁵ “Pastoralist and Farmer-Herder Conflicts in the Sahel | Climate-Diplomacy,” January 1, 1944, <https://climate-diplomacy.org/case-studies/pastoralist-and-farmer-herder-conflicts-sahel>.

⁸⁶ “Kenya 2022 Drought Response in Review - Kenya | ReliefWeb,” February 23, 2023, <https://reliefweb.int/report/kenya/kenya-2022-drought-response-review>.

across Africa, and Kenya is no exception, due to the high dependency of its residents on pastoralism. Pastoralism is one of the foundations of Kenya's economy, employing over 9 million people in Kenya.⁸⁷ For many years, transhumant pastoralists, farmers, and residents have stably coexisted across this semi-arid landscape. However, recently political unrest and limited resources have strained tensions among the pastoral community.

Before British colonization of Kenya, pastoralism was mainly controlled by tribes competing for pasture and control of cattle. The Somali, Borana, Maasai, and Turkana tribes fought for control of Kenya's land, eliciting conflicts based on herd size, community, and access to pasture. These dynamics continued past the British control of Kenya and are still present today.⁸⁸ One aspect of pastoralism, cattle raiding, was previously seen as a way to balance the amount of livestock grazing on a limited amount of pasture, but has escalated into large scale, violent conflicts. Cattle raiding usually involves a group of armed, young men who use force and weapons to steal livestock herds from pastoralists.⁸⁹ These attacks are often prompted by political incentives, land use issues, or limited access to resources.⁹⁰ Surprisingly, these attacks have been found to occur at higher rates during periods of higher rainfall, due to the opportunistic mindset of cattle raiders.⁹¹ During periods of higher rainfall, there is more available pasture and more pastoralists available to be raided. While cattle raids have been mainly attributed to resource scarcity, they have also been tied to political violence.

⁸⁷ "Agriculture, Food and Water Security | Kenya," U.S. Agency for International Development, June 5, 2023, <https://www.usaid.gov/kenya/agriculture-food-and-water-security>.

⁸⁸ Richard Waller, "Pastoral Production in Colonial Kenya: Lessons from the Past?," *African Studies Review* 55, no. 2 (2012): 1–27.

⁸⁹ "Absorbing Climate Shocks and Easing Conflict in Kenya's Rift Valley | Crisis Group," accessed April 1, 2024, <https://www.crisisgroup.org/africa/east-and-southern-africa/kenya/b189-absorbing-climate-shocks-and-easing-conflict-kenyas-rift>.

⁹⁰ Ibid.

⁹¹ Karen M. Witsenburg and Wario R. Adano, "Of Rain and Raids: Violent Livestock Raiding in Northern Kenya," *Civil Wars* 11, no. 4 (December 1, 2009): 514–38, <https://doi.org/10.1080/13698240903403915>.

2.3.2 Pastoral Violence and Politics

Political violence has been a persistent issue throughout Kenya, notably intensifying during election periods, and has subsequently become intertwined with pastoral violence. In particular, cattle raiding is employed by groups as a strategy to enhance their economic standing and reinforce their ethnic bonds and political objectives.⁹² Ethnic factions and tribes aim to secure dominion over various Kenyan regions, with livestock raiding serving as a tactic to achieve such regional dominance.⁹³ Moreover, insufficient regulation of arms trading has allowed cattle raiders easier access to firearms, escalating the number of pastoral conflicts that culminate in violence.⁹⁴

2.3.2 Pastoral Violence and Land Ownership

While conflicts among pastoralists have only recently increased, these conflicts have been occurring since before colonialism. Before British colonization, Kenya's land was primarily controlled by local warlords, who took advantage of access to land and resources, causing instability even before colonization.⁹⁵ The British gained control of Kenya in the early 1900s and transformed northern Kenya into the Northern Frontier District (NFD). Since pastoralists migrate all across Kenya, the British government had a hard time controlling their movements. They focused their efforts on attempting to limit their movements between the NFD and Kenyan highlands.⁹⁶ The British government saw the NFD to be of no economic value to them and viewed the high concentration of Muslims to be a threat to the rest of the colony, so they attempted to confine pastoralists into the NFD. This limited the pastoralists' access to the

⁹² Roba Sharamo, "The Politics of Pastoral Violence: A Case Study of Isiolo County, Northern Kenya".

⁹³ Greiner, "Guns, Land, and Votes."

⁹⁴ Ibid.

⁹⁵ Saverio Krätli and Camilla Toulmin, "Farmer-Herder Conflict in Sub-Saharan Africa?"

⁹⁶ Saverio Krätli and Camilla Toulmin, "Farmer-Herder Conflict in Sub-Saharan Africa?"

highlands, which contained the wet pasture that they needed access to during the wet months. This also allowed the NFD to be relatively self-governed, preserving the local institutions and rules in the area.⁹⁷

Despite gaining independence in 1963, British impact still remained across Kenya. The 1970s and 80s brought about major droughts, causing many pastoralists to sell their cattle. The herds were often purchased by foreign large-scale cattle investors who employed local young men to manage their cattle for a low cost. Prices of cattle fell to nothing, because of the surplus of supply, and farmers began buying cattle as well. Many of these large scale cattle owners are still present today, and have more money and power than traditional pastoralists, increasing their access to key resources like land and water.

The recent increase in conservancies and the privatization of land has also increased tensions across Kenya.⁹⁸ In the 1970s and 80s, group ranches emerged as a result of overgrazing and land degradation. These group ranches were an attempt to give land access back to locals and created subplots for pastoralists to use to graze. However, this also had negative effects because subplots were sold to foreign investors, pushing many of the local tribes off their land they originally owned. These group ranches were also created as a way to reduce livestock numbers and contain pastoralists into a smaller area.⁹⁹ Group ranches, which were intended to solve issues of overgrazing and land degradation, marginalized certain groups and gave access and power to foreign investors and private landowners.

⁹⁷ Ibid.

⁹⁸ Conrad Schetter, Kennedy Mkutu, and Marie Müller-Koné, "Frontier NGOs: Conservancies, Control, and Violence in Northern Kenya," *World Development* 151 (March 1, 2022): 105735, <https://doi.org/10.1016/j.worlddev.2021.105735>.

⁹⁹ "Land Privatization and Pastoralist Well-being in Kenya - Lesorogol - 2008 - Development and Change - Wiley Online Library," accessed April 1, 2024, <https://onlinelibrary.wiley.com/doi/full/10.1111/j.1467-7660.2007.00481.x>.

Conservancies are another way in which tensions involving pastoralists escalated. Conservancies, which were originally intended to preserve wildlife and the landscape, alienated local tribes from their land and increased the amount of privately owned land in Kenya.¹⁰⁰ The concentration of privately owned land in the hands of wealthy landowners, was another way in which local Kenyan tribes were harmed.

Currently, conservancies, group ranches, and other privately owned land populate Kenya's landscape. These privately owned properties severely limit the access to land that the local tribes once had. In a country which is so heavily dependent on scarce resources, access to land becomes a primary concern for many Kenyans. Additionally, the historical legacy of colonialism is still present in Kenya and has resulted in the marginalization and alienation of many pastoralists. While colonialism brought a resurgence of privately owned land, recent droughts have prompted an increase in conservation efforts in the form of protected areas. The next section will explore protected areas and their involvement in pastoral violence.

2.3.4 Pastoral Violence and Protected Areas

Recently, certain types of protected areas, like community conservancies and reserves, have become enmeshed in pastoral violence in Kenya. Due to the land distribution of these protected areas which privatize a significant portion of the pastoral commons, they often limit pastoral migration corridors. Private land owners either ask pastoralists to pay a grazing fee and offer them temporary access to the land, or they completely refuse them the right to graze on the

¹⁰⁰ "Trans-Local Peace Building among Pastoralist Communities in Kenya-Case of the Laikipia Peace Caravan," accessed December 3, 2023, https://www.researchgate.net/publication/257589456_Trans-local_Peace_Building_among_Pastoralist_Communities_in_Kenya-Case_of_the_Laikipia_Peace_Caravan.

land which was previously free range land.¹⁰¹ The cadastral process of land partitioning, often present in reserves and some community conservancies, has been found to be suitable for sedentary livelihoods, like agriculture, but not for nomadic livelihoods, like pastoralism.¹⁰²

In response to community conservancies privatizing the pastoral commons, there have recently been attempts to integrate pastoralists into the tourism industry of conservancies. However, this transition resulted in many trade-offs for pastoralists. While tourism provides pastoralists with a stable, year-round income, pastoralists' overall economic outcome was found to decrease.¹⁰³ Moreover, integrating pastoralists into the tourism sector of conservancies gives conservancies primary control of their herds, limiting their migration patterns.¹⁰⁴

The incorporation of pastoralists into community conservancies has had mixed levels of success. In some cases, community members have integrated smoothly into the community conservancy model, resulting in an increase in their economic output and economic stability. However, this smooth transition is a result of community conservancies respecting the culture and traditions of pastoral communities and giving them power in the decision making process.¹⁰⁵ On the other hand, giving conservancies control of the pastoral commons has resulted in further dividing pastoral communities from community conservancies, inciting political violence and increasing land use tensions.

¹⁰¹M. Lengoiboni, P. van der Molen, and A. K. Bregt, "Pastoralism within the Cadastral System: Seasonal Interactions and Access Agreements between Pastoralists and Non-Pastoralists in Northern Kenya," *Journal of Arid Environments* 75, no. 5 (May 1, 2011): 477–86, <https://doi.org/10.1016/j.jaridenv.2010.12.011>.

¹⁰² Monica Lengoiboni, Arnold K. Bregt, and Paul van der Molen, "Pastoralism within Land Administration in Kenya—The Missing Link," *Land Use Policy*, Forest transitions, 27, no. 2 (April 1, 2010): 579–88, <https://doi.org/10.1016/j.landusepol.2009.07.013>.

¹⁰³ Claire Bedelian and Joseph O. Ogutu, "Trade-Offs for Climate-Resilient Pastoral Livelihoods in Wildlife Conservancies in the Mara Ecosystem, Kenya," *Pastoralism* 7, no. 1 (May 23, 2017): 10, <https://doi.org/10.1186/s13570-017-0085-1>.

¹⁰⁴ Ibid.

¹⁰⁵ Ken Ogao Oburah, Clement Lenachuru, and Wilfred O. Odadi, "Does the Community Conservancy Model Work for Pastoralists? Insights from Naibunga Conservancy in Northern Kenya," *Sustainability* 13, no. 14 (January 2021): 7772, <https://doi.org/10.3390/su13147772>.

Overall, protected areas in Kenya have had both positive and negative effects on pastoral communities in Kenya. However, with proper implementation and support offered to pastoral communities, there is a possibility for a harmonious relationship to be created. This thesis will attempt to gain a better understanding of what factors are contributing to pastoral violence and will hopefully offer solutions geared towards reducing future conflicts.

2.4 Recent Empirical Work and Hypotheses

While research has shown that climate change and drought increase conflict across the world, there has been little causal evidence found in Kenya. In Africa, Nunn and McGuirk (2021) found that droughts in areas with high levels of pastoralism led to an increase in conflict.¹⁰⁶ Hendrix et al. (2012) also found that rainfall variability had an effect on large scale and small scale conflicts in Africa. Despite this evidence, there have been regional studies conducted in Northern Kenya which found a causal relationship between higher rainfall and more conflict.¹⁰⁷ In this thesis, I will explore the relationship between climate change and conflict at the national level, which is a scope that hasn't been previously analyzed.

Additionally, I will be exploring how protected areas, a creation that came about due to climate change and human interaction with the environment, is a cause for conflicts in Kenya. Scholars have researched how protected areas impact human-wildlife conflicts due to their close proximity with neighboring communities.¹⁰⁸ However, another reason why conflicts occur in protected areas is their encroachment on pastoral rangelands.¹⁰⁹ Climate change has drastically

¹⁰⁶ McGuirk and Nunn, "Transhumant Pastoralism, Climate Change and Conflict in Africa," n.d.

¹⁰⁷ Witsenburg and Adano, "Of Rain and Raids."

¹⁰⁸ Moses Makonjio Okello and John Warui Kiringe, "Threats to Biodiversity and Their Implications in Protected and Adjacent Dispersal Areas of Kenya," *Journal of Sustainable Tourism* 12, no. 1 (January 1, 2004): 55–69, <https://doi.org/10.1080/09669580408667224>.

¹⁰⁹ "Wildlife-Community Conflicts in Conservation Areas in Kenya | African Journal on Conflict Resolution," accessed April 1, 2024, <https://www.ajol.info/index.php/ajcr/article/view/63311>.

reduced the availability of fresh grazing lands, leaving numerous pastoralists struggling to find adequate pasture. Given that protected areas occupy large portions of Kenya's terrain and frequently include areas of fresh pasture, there has been a recent uptick in tensions between pastoralists and the management of these protected zones. Consequently, my research will focus on examining conflicts within protected areas from the perspective of pastoralists, underscoring the role that the encroachment of protected areas into pastoral lands plays in exacerbating conflicts. This inquiry has guided me to the core question of my research: How is pastoral violence linked with protected areas within the broader context of climate change and conflict? In the following section, I present my hypotheses to explore this question.

Summary of Hypotheses	
Climate Change and Conflict	<i>Higher</i> effects of climate change will be correlated with <i>higher</i> conflict levels.
Protected Areas and Conflict	<i>Community oriented</i> protected areas and <i>reserves</i> will have the highest levels of conflict.
Protected Areas and Phytomass	Protected areas will have <i>higher</i> phytomass levels than other areas.
Climate Change and Pastoral Violence	<i>Lower</i> effects of climate change will be correlated with <i>higher</i> pastoral violence.
Protected Areas and Pastoral Violence	<i>Community oriented</i> protected areas and <i>reserves</i> will have the highest levels of pastoral violence.

This thesis aims to expand on the existing body of work regarding climate change and conflict, with a specific focus on understanding the roles of protected areas and pastoralism within this dynamic. My initial hypothesis posits that the environmental impacts of climate change, such as reduced rainfall and increased drought intensity, will correlate with heightened levels of conflict. This premise is supported by existing research that links significant climate

change impacts to increased conflict rates across Africa. I anticipate that this correlation will hold true at the national level in Kenya as well.

My second hypothesis suggests that community-focused protected areas, including community conservancies and community nature reserves, alongside other types of reserves, are likely to experience high conflict rates. This expectation is based on the susceptibility of community-managed lands to disputes over ownership and elevated instances of pastoral violence. Additionally, the limited human activity permitted in reserves may lead to pastoralists infringing on these lands, escalating boundary disputes.

Regarding phytomass within protected areas, I predict that these regions will exhibit higher phytomass levels compared to unprotected areas, thereby attracting pastoralists to these lands for grazing. Furthermore, I anticipate a connection between pastoral violence and less severe climate change effects, such as increased rainfall. This hypothesis is inspired by studies on livestock raiding in Kenya, which identified a correlation between higher rainfall and more frequent conflicts. I expect to observe a consistent pattern in my analysis. Lastly, I believe that community-focused protected areas and reserves will witness higher levels of pastoral conflict in comparison to other protected areas, primarily due to the absence of clear physical boundaries and encroachments into pastoral territories. In the following section, I will detail the methodology and statistical analyses planned for this thesis.

Chapter 3: Methodology

3.1 Data Sets

3.1.1 Conflict Data

In this paper, I will be using three different conflict data sets to analyze spatial trends in conflicts across Kenya which all contain data on individual reported conflicts. The first data set which I will be using is from the Armed Conflict Location & Event Data Project (ACLED) which contains data on “political violence, demonstrations (rioting and protesting), and other select non-violent, politically important events.”¹¹⁰ The ACLED data set contains 12,850 conflicts from 1997 to 2024 which took place in Kenya. These conflicts are also event based and are reviewed multiple times before being published in a data set.¹¹¹ I chose to use the ACLED data set in this thesis because it had a significant amount of recorded conflicts and because the data set specified the specific actors which were involved in each conflict.

The ACLED data set contains many key elements like date, actors, associated actors, latitude, longitude, the source the conflict was acquired from, and deaths. It also contains some elements that are particularly useful like type of violence and a more specific definition of the actors involved. For example, ACLED classifies the type of violence as “political violence”, “strategic violence”, or “demonstration” which would be useful when looking at the impact of politics on conflicts in Kenya. The data set also specifically defines the actors involved in the conflict by writing “Pastoralists (Kenya)” or “Farmers (Kenya)” in the actors column. For

¹¹⁰ “ACLED | Bringing Clarity to Crisis,” ACLED, March 22, 2024, <https://acleddata.com/>.

¹¹¹ “Quick Guide to ACLED Data,” *ACLED* (blog), accessed April 1, 2024, <https://acleddata.com/resources/quick-guide-to-acled-data/>.

example, 769 of the 12,850 conflicts state that pastoralists are one of the two actors involved in the conflict.

The second data set which I will be using is from the Uppsala Conflict Data Program (UCDP) which contains data on organized violence and armed conflicts around the world. The data set which I will be using contains 1,126 conflicts ranging from the years of 1989 to 2022 that took place in Kenya.¹¹² In this data set, each conflict is classified as a given event which UCDP defines as: “the incidence of the use of armed force by an organized actor against another organized actor, or against civilians, resulting in at least 1 direct death in either the best, low or high estimate categories at a specific location and for a specific temporal duration.”¹¹³ While smaller in size, the UCDP dataset places an emphasis on political conflicts and has a different spatial range than the ACLED data set. I chose to include this data set because it added additional conflicts in a different region of Kenya which the ACLED data set didn’t account for.

Some of the key information provided in the UCDP data set, which is similar to that of the ACLED data set, includes date, key actors on each side, source article, latitude, longitude, and deaths. In terms of pastoral conflicts, there are around 25 of the 1,126 conflicts which contain “herder”, or “pastoralist” which indicates a relatively low number of conflicts involving pastoralists. However, since these conflict events are mainly based off of news articles, the low number of farmer herder conflicts could indicate a selection bias geared towards political violence rather than pastoral violence or a lack of defining the types of conflict and the specific actors involved.

¹¹² “UCDP - Uppsala Conflict Data Program,” accessed April 1, 2024, <https://ucdp.uu.se/>.

¹¹³ Ralph Sundberg and Erik Melander, “Introducing the UCDP Georeferenced Event Dataset,” *Journal of Peace Research* 50, no. 4 (July 1, 2013): 523–32, <https://doi.org/10.1177/0022343313484347>.

The final data set being analyzed is from the Northern Rangelands Trust (NRT) which is a conservancy organization which manages 33 conservancies and 460,000 pastoral communities.¹¹⁴ NRT was mentioned in the literature review as well, however their main goal is to “to build resilient community conservancies, transform lives, and secure peace and conserving natural resources.”¹¹⁵ The data set which I will be using from NRT contains 953 total conflicts which were recorded between 2017 and 2022. These conflicts were all reported by or occurred near a conservancy, because NRT’s focus is mainly on conservancies and protected areas. Additionally, this conflict data set contains information on both interventions and conflicts. It also contains data on the conservancy which reported the conflict, latitude, longitude, deaths, type of intervention, and date. For the NRT data set, 168 of the recorded conflicts in the data set involved pastoralism or herders. Given this thesis’ focus on protected areas, the NRT data set was especially useful when analyzing the relationship between conflicts and protected areas, which is why I chose to include it.

Due to the differences in data set coverage, selection bias, and recorded information on each conflict, I chose to use all three of these data sets to analyze conflicts in Kenya. Even though the three conflict data sets together do not account for all conflicts across Kenya, they do allow for three different perspectives on conflict in Kenya and will hopefully allow for more robust findings. The ACLED data set allows for a large-scale, more comprehensive conflict analysis, while the UCDP adds many instances of additional conflicts involved in political violence. Finally, the NRT data set accounts for conflicts near or within protected areas, which is important for this thesis specifically. In this thesis, I will be analyzing all three data sets separately to see whether the trends I’m analyzing are present across all three data sets.

¹¹⁴ “Northern Rangelands Trust | KWCA.”

¹¹⁵ Ibid.

3.1.2 Climate Change Data

Climate change data is another type of data which I will be using for my analysis. Because there are a variety of effects that climate change has on the environment¹¹⁶, I won't be able to fully capture the effects of climate change on conflict. Due to this, I selected a few variables which measure the effects of climate change on the atmosphere, which include precipitation, drought, normalized difference vegetation index (NDVI), and phytomass.

To begin the analysis, I acquired annual precipitation data from the World Bank, in the Climate Change Knowledge Portal, which is a resource which contains data on various factors on countries around the world.¹¹⁷ The World Bank has multiple climate change data collection centers scattered throughout Kenya, which they take the average recorded values from.¹¹⁸ The data I chose is annual rainfall data for each year which is averaged across Kenya and recorded in millimeters per year.¹¹⁹ In addition to the rainfall data, I also wanted to measure the effects of drought on Kenya's environment. I wasn't able to find a specific drought index which was measured annually, so I created my own index to measure the impacts of drought on Kenya. While precipitation and drought are correlated, drought is usually represented as the long term effects of a lack of precipitation¹²⁰, so I wanted to measure the long term effects of low precipitation levels on conflict using the drought index. To create the drought index, I used a variety of different sources, such as news articles and policy reports, to measure the impact of drought on Kenya as a categorical variable(see appendix 1 for sources). The drought index is

¹¹⁶ "Climate Change | United Nations."

¹¹⁷ "World Bank Climate Change Knowledge Portal," accessed April 1, 2024, <https://climateknowledgeportal.worldbank.org/>.

¹¹⁸ Ibid.

¹¹⁹ Ibid.

¹²⁰ "Temperature & Precipitation | Drought.Gov," accessed April 1, 2024, <https://www.drought.gov/topics/temperature-precipitation>.

measured as high, medium, or low, with high representing high levels of drought, medium representing medium levels of drought, and low representing low levels of drought.

The next climate data which I acquired was NDVI, which is a measurement of vegetation greenness and plant health. NDVI is calculated using satellite imagery which takes the amount of near-infrared light divided by the amount of red visible light, which leaves the amount of green light or greenness in an area.¹²¹ This will help give another measurement of the effects of climate change other than precipitation and drought. In addition, NDVI is correlated with pastoralism because pastoralists depend on healthy and green pastures in order for their herds to stay alive and healthy.¹²²

I acquired the NDVI data from Humanitarian Data Exchange which used NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) collection 6.1 from the Aqua and Terra satellite aggregated by sub-national administrative units to create annual NDVI levels across Kenya during the years of recorded conflicts.¹²³ The data set also had monthly and 10 day averaged data. I chose to use annual data for the climate change variables which I am analyzing because I wanted to measure the long term effects of climate change on conflict rather than short term effects. Usually, the direct effects of climate change, like low precipitation, NDVI, or drought levels, take significant time to impact conflict, especially pastoral ones, due to the indirect pathways involved in the climate and conflict dynamic.¹²⁴

In addition to a temporal analysis, I also conducted a spatial analysis of rainfall and phytomass levels and conflict. For this section, in addition to these conflict data sets, I will also

¹²¹“NDVI FAQs: Frequently Asked Questions About The Index,” August 30, 2019, <https://eos.com/blog/ndvi-faq-all-you-need-to-know-about-ndvi/>.

¹²² Galaty, “Pastoralism in Anthropology.”

¹²³ “Kenya: NDVI at Subnational Level - Humanitarian Data Exchange,” accessed April 1, 2024, <https://data.humdata.org/dataset/ken-ndvi-subnational>.

¹²⁴ Koubi, “Climate Change and Conflict.”

be using a data set from Nathan Nunn and Eoin McGuirk which was used in their paper “Transhumant Pastoralism, Climate Change and Conflict in Africa.”¹²⁵ Nunn and McGuirk analyze pastoral communities and the impact of droughts on conflicts in neighboring communities. Their main findings in their data is that drought in areas dependent on pastoralism lead to conflicts in nearby areas.¹²⁶ The data set that they use contains a combination of conflicts from UCDP and ACLED in addition to a measure of land suitability for pastoralism which they derive from information on ethnographic research on pastoralism that has already been done.¹²⁷

For their analysis, Nunn and McGuirk created a 0.5 degree cell grid across Africa where they matched each conflict with the nearest grid cell. For each grid cell, they assigned an annual value of many different variables, like dependence on pastoralism as a livelihood, phytomass, and rainfall. For the climate change and conflict chapter of this thesis, I will be using their regional values of phytomass and rainfall for each grid cell. By averaging the phytomass and rainfall values for each cell, I will map the areas with the highest phytomass levels and rainfall to see how it intersects with conflicts.

3.1.3 Protected Area Data

In addition to the data sets and climate change data mentioned above, I will also be using spatial boundary data on national protected areas in Kenya. I acquired this data from Protected Planet which is an open source data platform on protected areas across the world.¹²⁸ It contains spatial data on conservancies, forest reserves, national parks, group ranches, and other protected areas in Kenya.¹²⁹ The data, which is in a shapefile format, contains latitude and longitude

¹²⁵ Eoin F McGuirk and Nathan Nunn, “Transhumant Pastoralism, Climate Change and Conflict in Africa”.

¹²⁶ Ibid.

¹²⁷ Ibid.

¹²⁸ “Protected Planet | Kenya,” Protected Planet, accessed April 1, 2024, <https://www.protectedplanet.net/en/country/KEN>.

¹²⁹ Ibid.

coordinates of the current boundaries of protected areas across Kenya. I will use this data to count the number and type of conflicts which are occurring within protected areas.

3.1.4 Pastoralism Conflict Data

To measure pastoral violence, I will be manipulating a few of the data sets which I will be using for my climate change and conflict analysis. The ACLED data, which contains information on the actors involved in each conflict, clearly states whether pastoralists are one of the actors involved in each conflict. Therefore, using a filter, I was able to create a new data set which only included ACLED conflicts which had pastoralists as one of the actors involved in each conflict. This left 769 individual conflicts which range from 2001 to 2023.

Additionally, the data set from Nunn and McGuirk included a pastoralism index which indicated an area's dependence on pastoralism as a livelihood. A pastoralism index was assigned to each 0.5 degree grid cell. A high pastoralism index, closer to 1, indicates that the specific 0.5 degree grid cell has a higher concentration of its population which relies on pastoralism and a low index, closer to 0, indicates that the grid cell has a lower concentration of its population which relies on pastoralism or agriculture. I will be using the values of the pastoralism index for each grid cell to evaluate the locations in which pastoralism is the most widely used as a major livelihood in order to properly understand the dynamics of pastoral conflicts.

3.2 Main Variables

Based on the literature review and analysis of recent empirical work, for this thesis I will be analyzing an indirect pathway linking climate change to pastoral conflicts in Kenya, utilizing protected areas and as a mechanism connecting the two. Figure 1 shows the relationship of the main variables which I will be analyzing.

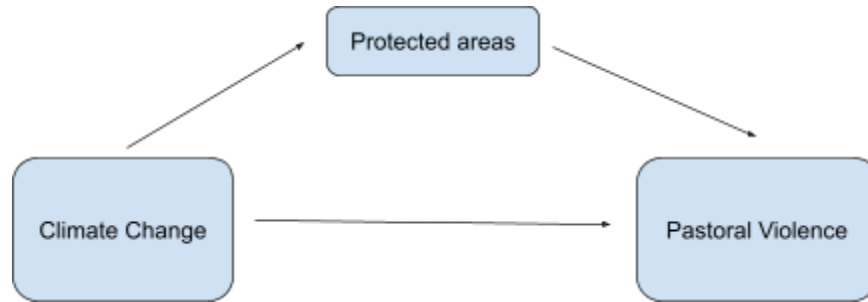


Figure 1 shows variables and relationships which will be analyzed in this paper. These are climate change, protected areas and pastoral conflicts.

The main goal of my data analysis is to analyze how these variables relate to each other, primarily focusing on how protected areas are intertwined in the climate change and conflict relationship. I will be using statistical tests and geospatial analysis using QGIS modeling to explore the interactions between these variables, which will be outlined in the next section.

3.3 Statistical Analysis Methods

The first dynamic which I will analyze in this thesis is the relationship between climate change and conflict. I will do this by conducting a linear regression comparing the amount of conflicts with various measurements of the impact of climate change on Kenya. The measurements which I will be using are precipitation levels, drought severity, and NDVI which are all annual measurements. I opted to examine these variables on an annual basis to assess the extended effects of climate change on conflict. Given the complexity and the multitude of factors at play in the climate change-conflict nexus, the immediate repercussions of climate change on conflict might not be readily apparent. A shorter analysis time frame might not sufficiently capture the development of conflicts, as it may take a significant period for these impacts to become evident.

To compile these climate measurements with the conflict data sets, I grouped each data set by year and found the number of recorded conflicts for each year. Next, I ran a linear regression model on the number of recorded conflicts and the annual climate change variable to test for statistical significance. In the linear regression model, I added a binary variable of election year, which is 1 for years where an election occurred and 0 for years when an election didn't occur. This is to account for any effect that an election may have on the number of conflicts recorded in a given year. In addition, I created an interaction term, which is included in the model as the climate change variable multiplied by the election year variable. The interaction term will help identify the impact of climate change on conflict during election years to ensure that election years aren't a confounding variable driving conflict. The results from this analysis will be outlined in Chapter 4 where I will run the linear regression model on each data set to test for statistical significance.

Chapter 4 will also include a spatial analysis of temperature and phytomass levels and conflict. Using QGIS modeling, I found the average precipitation and phytomass levels for each 0.5 degree grid cell across Kenya and created a heat map where higher values correspond to darker shades on the map. I overlaid this map with the individual conflicts recorded for all three data sets to see how higher levels of rainfall and phytomass interact with conflicts. For Chapter 4, I hypothesize that lower levels of rainfall, higher levels of drought, lower levels of NDVI, and lower phytomass levels will be associated with higher conflict. This is because these variables all represent high impacts of climate change, and I hypothesize that higher impacts of climate change on Kenya's ecosystem will drive conflicts.

In Chapter 5, I will build upon this analysis by incorporating protected areas into the climate change conflict analysis. Starting out by mapping each data set's recorded conflicts onto

a map of Kenya to see whether there are any spatial disparities in data set coverage, I expect the data sets to cover different parts of Kenya due to the different focuses of each data set.

Additionally, this analysis will allow me to see areas where conflicts are more clustered and areas where conflicts are more sparsely located. After this, I will map all of the protected areas onto a map of Kenya to look at similar spatial patterns. The map will show the boundaries of all types of protected areas in Kenya, with certain colors corresponding to different designations of protected areas. I will also create a figure representing the count of each type of designation of protected areas to see how many of each designation is present within Kenya. Before the next section of the analysis it will be important to get an understanding of which types of protected areas are most common in Kenya.

Next, I will use analysis tools in QGIS to see how many conflicts from each data set occur within the boundaries of protected areas. This will help with gaining a better understanding of the coverage of the conflict data sets and the relationship between protected areas and conflict. Using QGIS, I can analyze how many points, or conflicts, are present inside polygons, or borders of conservancies. From there, QGIS outputs a new data frame which contains every protected area in Kenya and the number of conflicts which occurred within their boundaries. I also will analyze which type of protected areas had recorded conflicts occur within their boundaries. This will give a better understanding of which types of protected areas are closely related to ongoing conflicts. In addition, I calculated the average conflicts per hectare inside each protected area to account for size differences between protected areas. Since certain types of protected areas are much larger than others this will give a baseline to see how concentrated conflicts are in protected areas. I also calculated standard error for each designation to see whether the differences in conflicts per hectare were significant for certain designations.

For the final chapter of the results section, I will focus on the impact of pastoralism and pastoral violence on the conflict and protected area relationship. I will begin this by analyzing the average phytomass values inside protected areas compared to outside protected areas. Using QGIS, I will find the average phytomass values from the 0.5 degree grid cells which fall inside protected areas and the average phytomass values of the 0.5 degree grid cells which fall outside protected areas. The calculation of these values will allow me to conduct a Welch two sample t-test between these two averages to test for statistical significance. Statistical significance will indicate a relationship between higher phytomass levels and the presence of a protected area. If these values are statistically significant, it will create a possible pathway for conflict. This is because pastoralists prefer to graze on land with higher phytomass levels, and if areas inside protected areas have higher phytomass levels, then pastoralists have an incentive to use protected area land to graze, which could incite conflicts.

In the next section, I will also be using the data set from Nunn and McGuirk, focusing on their pastoralism index, which indicates an area's dependence on pastoralism as a livelihood. Using QGIS, I created a heat map where areas with a higher dependence on pastoralism are darker to see how pastoralism intersects with protected areas. If there is higher dependence on pastoralism levels near protected areas, then it gives pastoralists another mechanism to use protected area land to graze on.

In the final section of this chapter, I will subset the ACLED data set to isolate pastoral conflicts apart from general conflicts. This will leave around 769 conflicts, all of which include pastoralists as one of the actors involved in conflicts. I will be using this new pastoral data set to test trends in pastoral violence over time and across Kenya. To begin, I mapped pastoral conflicts in Kenya, but corresponded the color of each conflict to a year. This will help understand where

pastoral violence is concentrated and how these conflicts have evolved spatially over time. After this, I found the percentage of conflicts which involved pastoralists compared to the total number of recorded conflicts and analyzed how it evolved over time. This data allowed me to conduct a similar linear regression model to that of the first results chapter, however, in this instance I used pastoral conflicts rather than the total number of conflicts. I ran a linear regression on the number of pastoral conflicts with average annual rainfall to test for statistical significance. I originally hypothesized that higher rainfall will be correlated with higher pastoral conflicts, due to the recent increase in cattle raiding, which mainly occurs during periods of high rainfall. Finally, I used QGIS to see how many pastoral conflicts were occurring within the boundaries of protected areas and what kind of designation of protected areas had the most recorded conflicts. This will give a better sense of the relationship between pastoral violence and protected areas and which designation of protected areas has the highest number of recorded conflicts. In the next chapter, I will begin the results section, which focuses on the relationship between different climate change variables and conflicts.

Chapter 4: Climate Change and Conflict

4.1 Annual Rainfall and Conflict

As stated in the literature review, climate and conflict are two variables which have been found to be linked in a variety of different settings. Some researchers viewed climate change as a main variable in predicting conflict, while others saw it as one of many variables involved in violence. In this chapter, I plan to analyze whether climate change and its effects play an instrumental role in creating conflicts. I plan on using a similar methodology to that of the researchers mentioned in the literature review, which use drought indices, like precipitation and forest coverage, to measure the correlation with the number of conflicts. While this method can not prove causation between drought and conflicts, it can suggest a correlation between the two variables which will assist with a deeper analysis in the future chapters.

To begin, I analyzed the relationship between the number of recorded conflicts and annual rainfall levels in Kenya. I acquired annual rainfall data from the World Bank, a platform which contains a variety of data sets on countries around the world. The World Bank compiled rainfall data from different areas across Kenya and took the average rainfall levels per year. My hypothesis states that higher amounts of recorded conflicts will be correlated with lower annual rainfall levels, indicating that a larger impact of climate change should be correlated with significantly higher conflicts.

For the first portion of the analysis, I ran a linear regression to test statistical significance between the number of conflicts recorded per year and the average rainfall per year. I used all three data sets, including the ACLED, NRT, and UCDP data sets. I kept the analysis separate for

each data set to account for any differences in coverage between data sets. Additionally, due to the impact that politics and election years have on violence, I used election years as a covariate to control for any effects of election violence on the relationship between climate change and conflict.¹³⁰ I also created an interaction variable between election year and annual rainfall to test whether the relationship between precipitation and conflict is stronger or weaker during an election year. Since the NRT data set only covers the years of 2017 to 2022, which included only 2 election years, I only made the interaction variable for the ACLED and UCDP data set. This relationship is presented in Table 1, which shows the linear regression between number of conflicts and annual rainfall levels for each data set. Rainfall levels were analyzed in millimeters per year.

Table 1: Annual Rainfall and Conflict Regression Results Summary

	Estimate	Std. Error	t value	Pr(> t)
Model 1: ACLED				
(Intercept)	-244.8883	354.8183	-0.69	0.4973
Annual Rainfall	0.7498	0.4759	1.58	0.1294
Election Year	-224.3631	995.7984	-0.23	0.8238
Annual Rainfall:Election Year	0.5816	1.1484	0.51	0.6176
Model 2: UCDP				
(Intercept)	-22.3793	30.6806	-0.73	0.4714
Annual Rainfall	0.0744	0.0420	1.77	0.0871.
Election Year	-3.0525	75.6924	-0.04	0.9681
Annual Rainfall:Election Year	0.0029	0.0908	0.03	0.9745
Model 3: NRT				
(Intercept)	39.2870	819.5072	0.05	0.9661
Annual Rainfall	0.1437	0.8493	0.17	0.8812
Election Year	73.4790	265.3062	0.28	0.8078

Note: P-values are reported in the right column. Significance codes: '***' p<0.001, '**' p<0.01, '*' p<0.05, and '.' p<0.1.

Table 1 shows the linear regression between the ACLED, UCDP, and NRT data sets and the annual rainfall data(in millimeters per year). The p values are interpreted in the note below and election years are also analyzed as a covariate. The ACLED and UCDP models also include an interaction term which represents the relationship between annual rainfall and election years on conflicts.

¹³⁰Kirsten Johnson et al., “A National Population-Based Assessment of 2007–2008 Election-Related Violence in Kenya,” *Conflict and Health* 8, no. 1 (January 18, 2014): 2, <https://doi.org/10.1186/1752-1505-8-2>.

The first data set which was analyzed in the regression table is the ACLED data set, which has around 12,800 recorded conflicts from the years of 1989 to 2023. All 12,800 conflicts were analyzed with annual rainfall data from those years as well. The linear regression shows a positive correlation between the number of recorded ACLED conflicts and higher annual rainfall. The analysis shows that, during years where an election did not occur, a one unit increase in the annual rainfall, in millimeters per year, was associated with a 0.74 unit increase in the number of recorded ACLED conflicts. This relationship is not quite statistically significant with a p value of 0.129, however the p value falling near 0.01 indicates a strong positive correlation between higher rainfall levels and more conflict. It is important to note that this finding goes against my original hypothesis, which was that lower rainfall would be correlated with higher conflict levels. Additionally, for the election year analysis, the p value was not significant for the election year term or for the interaction term, indicating that the presence of an election did not have a statistically significant difference on the number of conflicts in a year.

A similar trend was also found for the UCDP data set, which analyzed conflicts over the years of 1997 to 2023. A one unit increase in the average annual rainfall, during a non-election year, was correlated with a 0.07 unit increase in the number of recorded conflicts. This relationship was statistically significant with a p value of 0.0871 which also goes against my original hypothesis. Additionally, the UCDP analysis found no statistically significant relationship between the interaction term and the number of conflicts or between the presence of an election year and the number of conflicts. The third model was analyzed between the NRT data set and the annual rainfall levels. It is important to note that the NRT data set only contains conflict data between the years of 2017 and 2022, so these results are less applicable to the general trends I am analyzing across larger time periods. However, the regression found that a

one unit increase in annual rainfall was correlated with a 0.14 unit increase in the number of recorded conflicts in the NRT data set. This result had a p value of 0.88 which indicates a low correlation between rainfall and conflict.

Despite the third model's findings, the first two indicate a completely different relationship between conflict and climate change than predicted in my hypothesis. The results indicate a positive correlation between average annual rainfall and the number of recorded conflicts, indicating that climate change may not be directly contributing to an increase in conflicts. This suggests that there are many other factors that are most likely driving conflict, some of which I plan to discuss in the next few chapters of this thesis. For the next section of this chapter, I will analyze the relationship between drought and conflict. Drought is dependent on a variety of different factors, including rainfall, so given the previous findings in table 1, it is important to see whether these trends are also present for droughts across Kenya as well.

4.2 Annual Drought and Conflict

Other than the annual rainfall data which I acquired from the World Bank, there is limited cohesive data on annual drought levels in Kenya over the past 30 years. However, there is a surplus of news articles and research which documented droughts over the past century in Kenya. Due to this, I compiled various sources which contain drought data, including government research, news articles, and rainfall data to create an index which represents the severity of drought by year in Kenya, during the years with recorded conflicts(see appendix).

For the index, each year is divided into categories based on drought and precipitation level severity, which is organized as high, medium, or low. High indicates low precipitation levels and high drought severity, while medium indicates medium levels of both, and low

indicates high precipitation levels and low drought severity. These levels were created with respect to Kenya, so while low drought levels may be low with respect to Kenya, it may still be high with respect to the rest of the world. From there, I used the drought indicator of high, medium, or low as a categorical variable and counted the number of recorded conflicts for each data set by year. I ran a linear regression to assess the statistical significance of a one unit increase in the drought indicator with each year which is illustrated in Table 2. I also controlled for election years in this regression to control for any effect that election violence may have on the number of conflicts. Similar to the previous regression model, I used an interaction term of drought index times election year to see how drought levels interacted with conflicts when an election was present and when an election wasn't present. The linear regression results are shown in table 2.

Table 2: Annual Drought Levels and Conflict Regression Results Summary

	Estimate	Std. Error	t value	Pr(> t)
Model 1: ACLED				
(Intercept)	607.3732	209.8386	2.89	0.0084**
Drought	-132.4014	85.6662	-1.55	0.1365
Election Year	426.1268	447.6010	0.95	0.3514
Drought:Election Year	-114.3486	292.3968	-0.39	0.6995
Model 2: UCDP				
(Intercept)	62.2364	18.7232	3.32	0.0023***
Drought	-13.2091	7.5060	-1.76	0.0886.
Election Year	-18.8902	33.0473	-0.57	0.5718
Drought:Election Year	11.1706	17.4221	0.64	0.5263
Model 3: NRT				
(Intercept)	159.0000	364.0224	0.44	0.7049
Drought	14.0000	275.1751	0.05	0.9640
Election Year	74.0000	275.1751	0.27	0.8132

Note: P-values are reported in the right column. Significance codes: '***' p<0.001, '**' p<0.01, '*' p<0.05, and '.' p<0.1.

Table 2 shows the linear regression between the ACLED, UCDP, and NRT data sets and the drought index, which was assigned a numerical value of 1, 2, or 3.

The regression results, which are shown in table 2, show similar results to that of table 1, which analyzed conflict and annual rainfall. The drought index was assigned values to create a factor variable to assist with the regression, so high values were represented by a 3, medium were represented by a 2, and low values were represented by a 1. Therefore, a higher drought index would mean a higher number of the drought index. The first data set which was analyzed, ACLED, found a slightly negative correlation between conflict and drought. Model 1 shows that a one unit increase in the drought index, indicating higher drought levels, is correlated with a 132 unit decrease of the number of recorded ACLED conflicts. This indicates that a lower drought index, meaning lower visible impacts of climate change on the environment, is associated with a higher amount of conflict. This result had a p value of 0.13 which is not statistically significant however, it is relatively close to 0.1 indicating a negative correlation between drought levels and conflict. Similar to the previous model, election year or the interaction term didn't have a statistically significant relationship, suggesting that there is little correlation between election year and conflict for the ACLED data set.

For the second model, a similar result was found with a slightly higher p value of 0.086 for the UCDP data set. A one unit increase in the drought index was correlated with a 13.2 unit decrease in the number of recorded conflicts. This result was statistically significant with a p value of 0.086 suggesting a statistical correlation between lower drought levels and higher conflict. The results from the third model, with the NRT data set, were not statistically significant and had a high p value indicating no correlation. This could also be due to the limited coverage of the NRT data set, as it is only covering the years of 2017 to 2022. For both models, neither the election year nor the interaction term was significant.

The results of table 1 and table 2 indicate surprising findings, which present a relationship opposite of what was expected in the hypothesis. Both models suggested that higher rainfall and lower drought levels were associated with years with more recorded conflict. While this does not follow along with my hypothesis, it also suggests that the relationship between conflict and climate change is much more nuanced than I originally hypothesized. While I previously analyzed drought and rainfall, there are also many other variables which influence climate change. Due to this, for the next section, I will be analyzing NDVI to see how the number of conflicts is associated with phytomass across Kenya.

4.3 Annual NDVI and Conflict

For this section, I acquired data on the NDVI across Kenya which, as stated in the methodology, represents vegetation greenness and plant health. This data was found on the humanitarian data exchange website and contains NDVI values for every year from 2002 to 2023.¹³¹ These values were originally acquired from NASA's Moderate Resolution Imaging Spectroradiometer (MODIS) which uses satellite imaging to capture greenness and vegetation health. NDVI values fall between 0 and 1, with 1 indicating higher levels of greenness. I began by grouping the data on an annual level to keep the time value consistent for the entire chapter. As stated earlier, analyzing the relationship between climate and conflict on an annual level helps understand more general trends and gives time for conflict to react to changes in the climate. Therefore, I found the average NDVI by year for Kenya and ran a linear regression comparing it with conflict levels for each data set. The linear regression model is similar to the first two sections and uses NDVI instead of rainfall and drought levels. I also accounted for election years

¹³¹“Kenya: NDVI at Subnational Level - Humanitarian Data Exchange,”

in the model and created an interaction variable similar to the first two tables. The results of the linear regression are found in Table 3.

Table 3: NDVI and Conflict Regression Results Summary

	Estimate	Std. Error	t value	Pr(> t)
Model 1				
(Intercept)	-2005.1473	2039.8939	-0.98	0.3394
NDVI	4164.3159	3630.5696	1.15	0.2673
Election Year	11504.9270	4131.1152	2.78	0.0127*
NDVI:Election Year	-19599.6716	7342.1120	-2.67	0.0162*
Model 2				
(Intercept)	-95.9091	224.9401	-0.43	0.6752
NDVI	252.9605	400.3447	0.63	0.5359
Election Year	426.8760	455.5401	0.94	0.3618
NDVI:Election Year	-754.5679	809.6182	-0.93	0.3644
Model 3				
(Intercept)	957.9727	2822.9696	0.34	0.7667
NDVI	-1344.6520	4853.2080	-0.28	0.8077
Election Year	23.6430	311.5358	0.08	0.9464

Note: P-values are reported in the right column. Significance codes: '***' p<0.001, '**' p<0.01, '*' p<0.05.

Table 3 shows the linear regression between the ACLED, UCDP, and NRT data sets and NDVI values, which are between 0 and 1.

The results from Table 3 show no statistically significant relationship between NDVI and number of recorded conflicts. For the ACLED data set, a one unit increase in the NDVI value was associated with a 4,164 unit increase in the number of recorded ACLED conflicts. The p value associated with this is 0.26 which is not statistically significant. However, the election year covariate and the interaction term had statistically significant p values. The statistically significant p value means that the presence of an election year had a significant correlation with conflict levels. For the interaction term, the significant p value indicates that the effect of rainfall on conflict was stronger in election years. These results are interesting given that the election

covariate and interaction term were not significant for the previous models with the ACLED data set.

The results for the correlation between NDVI and conflict are similar for the UCDP data set as well which found a one unit increase in the NDVI to be associated with a 252 unit increase in the number of recorded UCDP conflicts. The p value was 0.53 which was similar to that of the first model, indicating no statistical significance. Finally, the NRT data set also had a high p value of 0.8 indicating that there was no correlation between NDVI and conflict. For both of these models, the election year covariate and the interaction term had no significant p values.

Overall, these three models indicate that there is no statistically significant relationship between vegetation levels and conflict for each data set. In the next section, I will conduct a spatial analysis of conflict, phytomass, and rainfall across Kenya. The previous analysis has looked at annual climate levels, however it is also important to do a spatial analysis in different areas of Kenya to see how robust this trend is.

4.4 Spatial Rainfall and Phytomass Levels and Conflict

In this section, I will conduct a spatial analysis of conflicts and rainfall levels across Kenya. Using a data set I acquired from Nunn and McGuirk in their paper on transhumant pastoralism and drought in Africa, I was able to find areas across Kenya which had higher rainfall levels and overlay them with the conflict data sets. Nunn and McGuirk arranged 0.5 degree grid cells across Africa which have individual conflicts associated with them. For each individual conflict, they have assigned a grid, temperature, phytomass level, rainfall, and a variety of other factors relating to the conflict. For this analysis, I took the average rainfall levels for each 0.5 degree grid cell across the 30 years which had recorded conflicts to see whether

conflicts were coinciding with areas of higher rainfall. This is mapped in Figure 2 which shows all recorded conflicts as yellow dots and areas with higher average rainfall, which appear as red areas on the map.

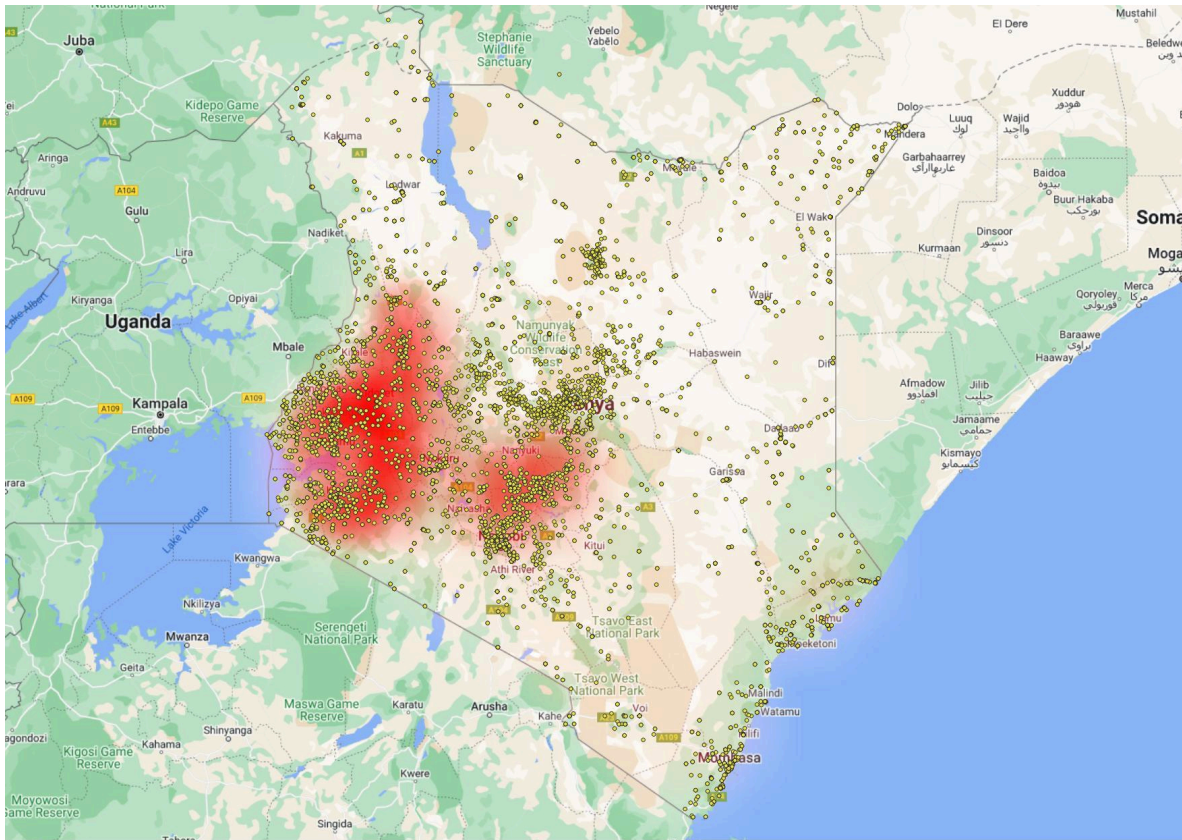


Figure 2 shows areas with higher average rainfall in red and recorded conflicts in yellow. The yellow dots represent conflicts from all three datasets.

Figure 2 shows the intersection of rainfall and conflicts in Kenya from 1985 to 2020. In the western and central parts of Kenya, rainfall and conflict coincide, where clusters of conflicts fall near or in areas with higher rainfall. This is true for many of the clusters in the western and central regions of Kenya. There is a smaller cluster of conflicts in the central northern part of Kenya which has lower rainfall levels, however this map mainly shows that there is a correspondence between areas with higher rainfall and conflict.

Next, using the data set from Nunn and McGuirk, I also mapped average phytomass levels from each 0.5 degree grid cell to see how they intersect with recorded conflicts. I used a similar methodology as the previous graph, taking the average phytomass level for each grid cell and mapping it using QGIS. Phytomass is defined as “any quantitative estimate of the total mass of plants in a stand, population, or within a given area, at a given time.”¹³² The results are shown in Figure 3, where the blue areas are cells with higher phytomass levels and the yellow dots represent individual conflicts.

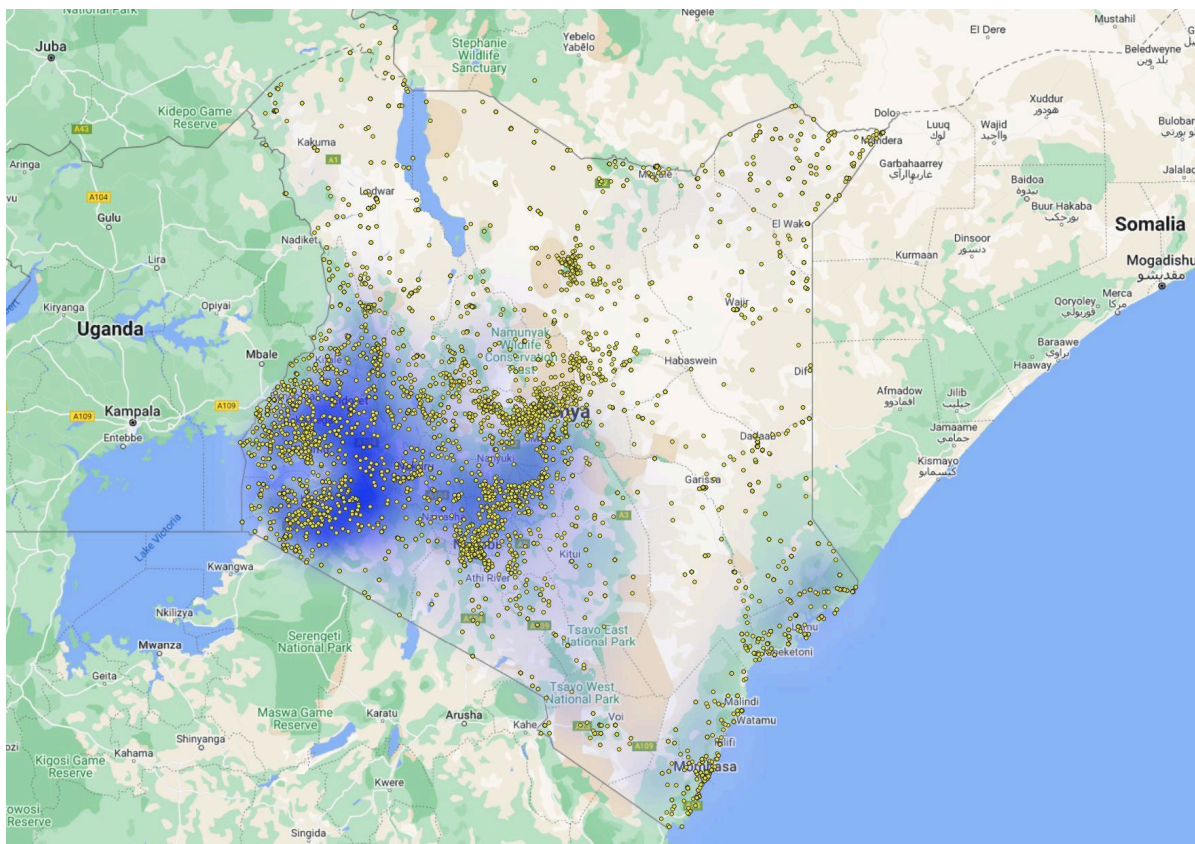


Figure 3 shows areas with higher average phytomass in blue. Each individual conflict is represented as a yellow dot.

Figure 3 shows a similar trend to Figure 2, where the western and central regions have higher phytomass levels and intersect with many of the clusters of conflict. The graph has a

¹³²“Phytomass — European Environment Agency,” Term, accessed April 1, 2024, <https://www.eea.europa.eu/help/glossary/gemet-environmental-thesaurus/phytomass>.

similar correlation, where higher phytomass levels seem to be occurring near clusters of conflicts. As stated earlier, there is also a cluster of conflicts in the northern central region which doesn't correspond to higher rainfall or phytomass, but the majority of other areas with concentrated conflicts have higher rainfall and phytomass levels.

I will discuss the implications of these results in the discussion section of this thesis, however these results mainly show mixed evidence linking climate change to conflict. For the first model, there was a correlation between average annual rainfall and annual conflict levels, indicating that more rain is associated with more conflict. This finding went against my original hypothesis which was that lower levels of rainfall would be associated with higher amounts of conflict. This hypothesis was also tested in the second section, which tested drought levels with conflict levels. I also found a statistically significant relationship between drought and conflict, indicating that higher levels of drought were associated with higher amounts of conflict.

The next portion of this chapter looked at NDVI and conflict levels. This model found that there were no statistically significant relationship between vegetation and conflict for any of the data sets. I also controlled for election year violence in the linear regression in order to ensure that there are no confounding factors. The final section of this chapter had similar results to the first two sections, but found that the positive correlation between rainfall and conflict was present over time and spatially in Kenya as well. This relationship was also true for phytomass levels, where areas with higher phytomass seemed to have a higher concentration of conflicts. Overall, these results show that the relationship between climate and conflict is much more complicated than I previously predicted. Due to the differing results of the regressions in this chapter, it is necessary to conduct other analyses to see what other factors are influencing this relationship as well. While I will not be able to examine all the factors that could be contributing

to climate change and conflict, I hope to understand a small portion of this relationship including protected areas, which I will be analyzing in the next chapter.

Chapter 5: Protected Areas and Conflicts

In the previous chapter, the relationship between climate change and conflict was analyzed, revealing surprising results suggesting that the dynamic is more nuanced than was previously hypothesized. Due to this, I plan on analyzing other variables which could be potentially involved in this relationship. Due to the recent increase in protected areas and pastoral violence, I will be analyzing protected areas and how they interact with climate change and conflicts in Kenya. Using the three conflict datasets which I used in the previous chapter, in this chapter I will be assessing how these conflicts interact with protected areas.

I will also be using protected area boundary data which I acquired from ProtectedPlanet to conduct a spatial analysis of the intersection between protected area boundaries and conflicts. This chapter outlines how many conflicts occurred inside the boundary of protected areas and what types of protected areas have the highest concentration of conflicts. As stated in my hypothesis, I don't expect conflicts to be concentrated inside protected areas, however I expect community based protected areas and reserves to have higher concentrations of conflicts.

5.1 Spatial Trends in Conflicts

In this chapter, I will focus on the relationship between protected areas and conflicts in Kenya, but it is first important to understand the spatial trends in the conflict datasets I am using in this thesis. Therefore, I mapped all three of the conflict datasets which I will be using which are NRT, ACLED, and UCDP in a QGIS modeling platform. In total, there were around 15,000 conflicts across the three data sets. Figure 4 shows the results of this analysis. Each dot represents an individual conflict. This figure helps understand the areas where conflicts are

concentrated and any spatial disparities in each of the data sets. The conflicts are color coded by data set.

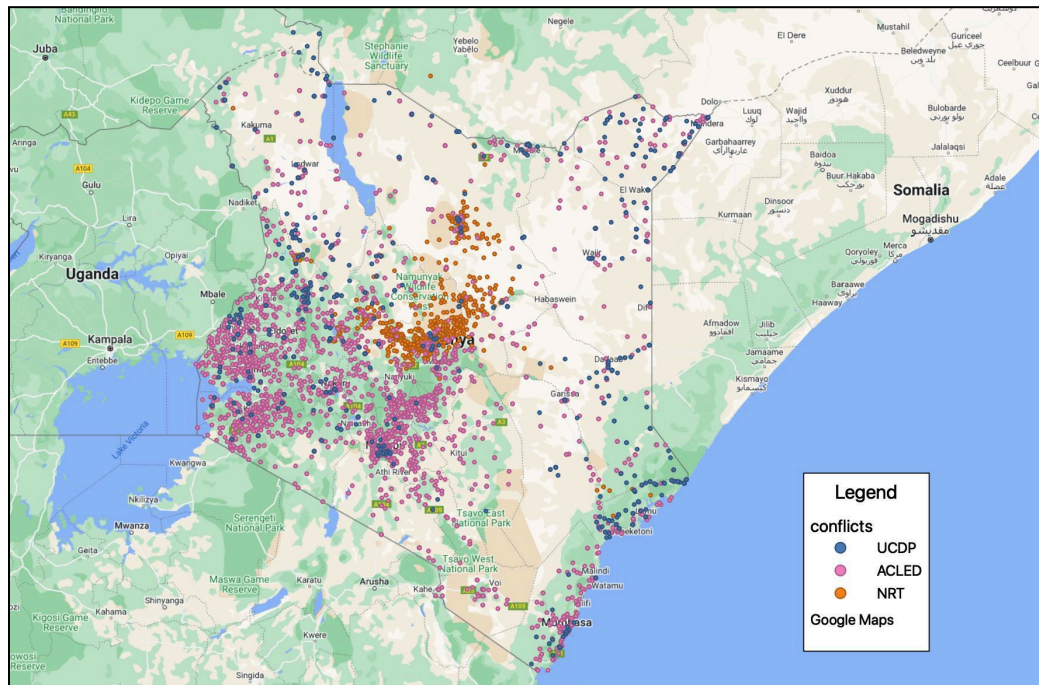


Figure 4 shows individual conflicts across Kenya for all three conflict datasets. The blue dots represent conflicts from the UCDP data set, the pink dots represent conflicts from the ACLED data set, and the orange dots represent conflicts from the NRT data set.

Figure 4 shows the spatial trends in the conflict data sets across Kenya. Overall, many conflicts tend to be concentrated in the southwest and central parts of Kenya. There also is a fair amount of conflict in the Nairobi area, as it is one of the largest cities in Africa and levels of violence are typically higher in cities. Other than the southwest and central regions, conflicts seem to be dispersed across the rest of Kenya. However, looking at the map, there are differences in data set coverage with different data sets having a higher concentration of recorded conflicts in certain areas.

The ACLED data appears to be clustered in the southwestern and central parts of Kenya. There is a large cluster of conflicts near Lake Victoria, near the Kenya-Uganda and

Kenya-Tanzania borders. Additionally, there are clusters of conflicts in the central regions near Nairobi. Other than this, there are additional conflicts spread out around the border, but not much coverage in the northern areas of Kenya. The UCDP data, which is represented as blue dots in Figure 4, don't seem to be concentrated in a singular area, but evenly dispersed across Kenya. There are also relatively less UCDP conflicts, around 1,100, so this could also account for the lack of clustering compared to the ACLED data set. There is a fair amount of UCDP conflicts in the northern areas of Kenya, which is useful due to the lack of northern coverage in the ACLED data set. Since Northern Kenya has high levels of pastoral violence, these conflicts will be a useful addition to the conflict datasets.

The NRT data is clustered around the middle sections of Kenya with a few conflicts reaching the northern areas as well. I believe that this is mainly attributed to NRT's geographical location which is protected areas in the central parts of Kenya. The NRT data will be especially useful when analyzing the relationship between protected areas and conflicts because they mainly record conflicts from conservancies and other protected areas in the central parts of Kenya. Overall, this spatial analysis will assist with future research on protected area, pastoralism, and conflict in the next sections of this chapter. Next, I will focus on the protected area boundary data and the spatial distribution of protected areas across Kenya.

5.2 Spatial Trends in Protected Area Boundaries

After analyzing the spatial trends of each conflict data set, I will be analyzing the spatial trends in protected areas. Using the boundary data which was acquired from ProtectedPlanet, I have mapped the protected areas within Kenya to get a better sense of which areas of Kenya are protected lands. Protected areas(PAs) have many different designations including community

conservancies, conservancies, national parks, marine parks, forest reserves, group ranches, and many more. Overall, Kenya has 409 protected areas, the majority of which are forest reserves, community nature reserves, and national parks.¹³³ Figure 5 shows the layout of all types of protected areas across Kenya.

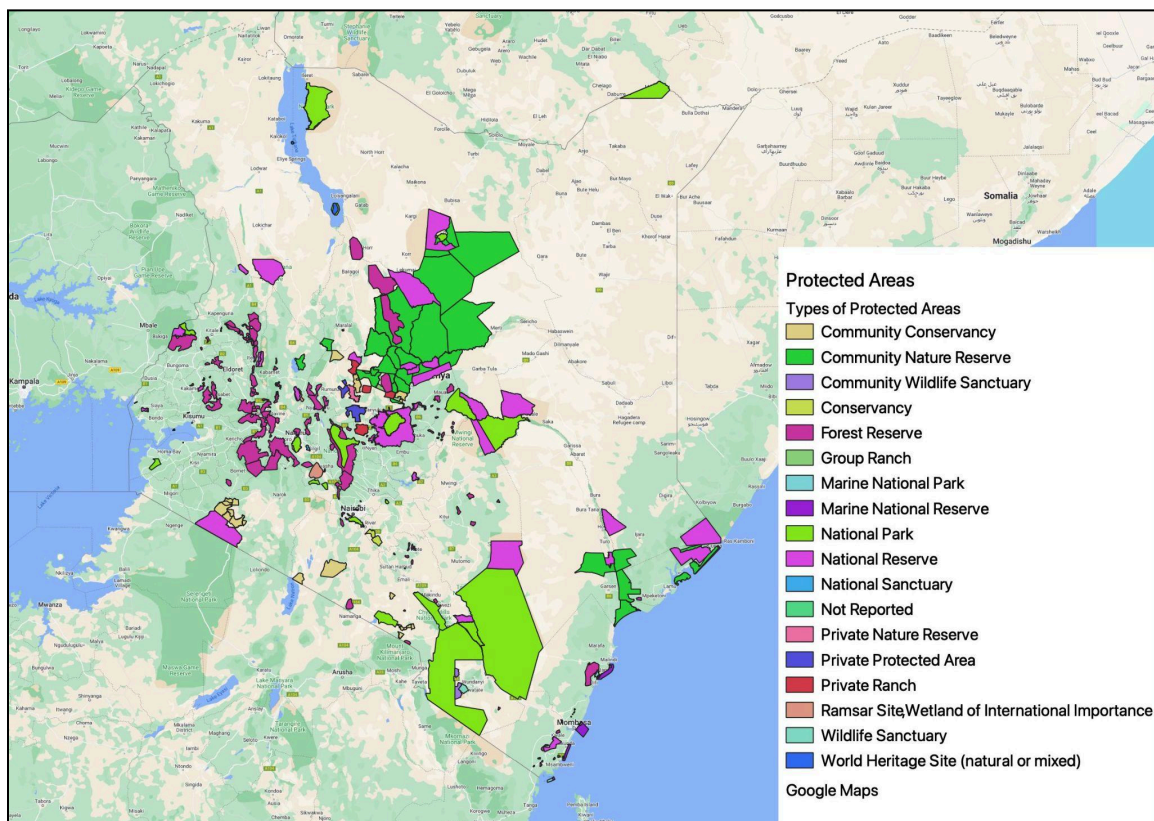


Figure 5 shows PAs across Kenya, with each color representing a different type of protected area. These designations were created by ProtectedPlanet.

Figure 5 shows the layout of protected areas across Kenya by designation, which were created by ProtectedPlanet. The figure shows that many of these protected areas are clustered in the central parts of Kenya, the majority of which are community nature reserves, national reserves, and national parks. While protected areas may seem like they cover a small portion of Kenya’s land, this map makes it clear that they cover a significant area and are distributed across

¹³³ “Protected Planet | Kenya.”

Kenya, indicating the importance of analyzing protected areas as a mechanism involved in the climate conflict relationship. The different colors represent the different types of protected area, which are outlined in the legend. The figure shows that community nature reserves and national parks have larger areas, while forest reserves and other designations have smaller areas and are more dispersed across Kenya. It is also important to note that the specific boundaries of protected areas across Kenya change frequently, due to conflicts over land ownership, so this data only represents boundaries during a specific time period. This data has been last updated in late 2023, however there is likely to have been changes in boundary data since the map has been created.

In addition to a geographical analysis of protected areas across Kenya, it is also important to consider the breakdown of the types of protected areas. Figure 6 shows how many of each type of protected area there are in Kenya. In the figure, I included the 5 types of protected areas which had the highest count. These categories include forest reserves, national reserves, community nature reserves, national parks, community conservancies, and private reserves.

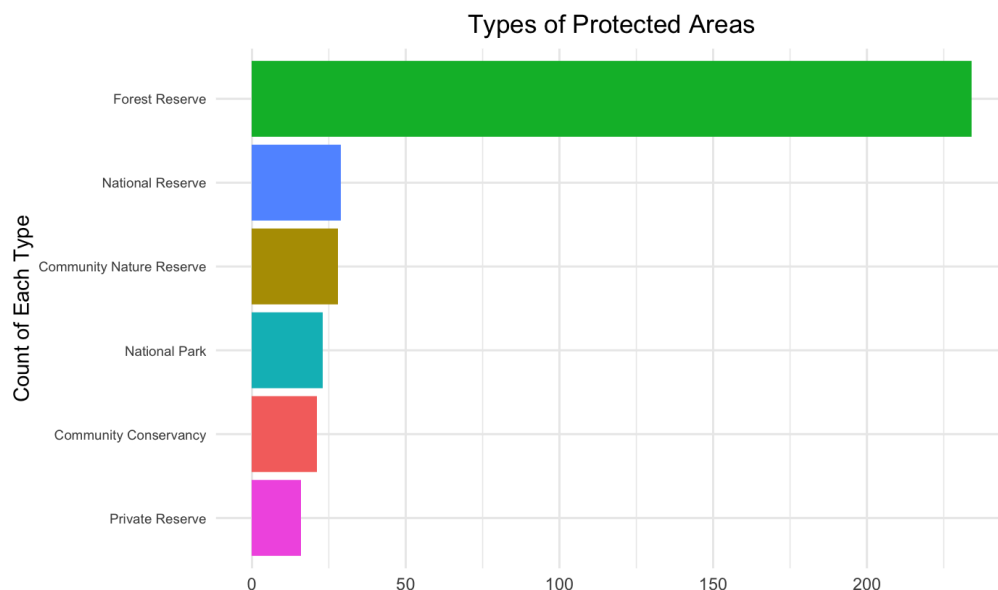


Figure 6 shows the count of each type of protected area in Kenya. Some designations are not graphed. The name of each designation is shown to the left of the graph.

Figure 6 shows that there are around 20 to 30 private reserves, national reserves, national parks, community nature reserves, and community conservancies across Kenya. There is also a large number of forest reserves, accounting for 234 which are present in Kenya. This figure shows that there is not only a large spatial coverage of protected areas across Kenya, but there are also large amounts of each protected area as well. Knowing which type of protected areas are the most present in Kenya will aid in future chapters which focus on conflict mitigation strategies in protected areas. In the next section, I will analyze the number of conflicts that occurred within different types of protected areas.

5.3 Conflicts Within Protected Areas

After analyzing the spatial trends in individual conflicts and protected areas across Kenya, I used QGIS modeling to calculate how many conflicts occurred inside protected areas. Using the latitude and longitude of each individual conflict, I was able to see whether each conflict occurred inside the boundaries of a protected area. QGIS calculated an attribute table which contained each protected area and how many conflicts were recorded inside each one. The attribute table was separated by data set and the total number of conflicts that occurred within protected areas and those that didn't are shown in Table 4. Table 4 represents the number of conflicts which occurred inside protected areas which have been separated by data set.

	ACLED	NRT	UCDP	Total
Conflicts inside PAs	824 (6.4%)	710 (74.5%)	140 (12.4%)	1674(11.2%)
Conflicts outside PAs	12026 (93.6%)	243 (25.5%)	986 (87.6%)	13255(88.8%)
Total Conflicts	12850	953	1126	14929

Table 4 shows the number of individual conflicts which have occurred inside the boundaries of a protected area for each data set(ACLED, NRT, and UCDP).

Table 4 shows the breakdown of conflicts which were recorded inside protected areas and conflicts which weren't recorded inside protected areas. For the ACLED data set, which has over 12,000 entries, there were only 6.4% of the reported conflicts which occurred inside protected areas. As modeled in Figure 3, the ACLED conflicts are clustered around Nairobi and the south western parts of Kenya near the Ugandan border which are areas with less land directed towards conservation and protected areas, which could explain the lower number of conflicts. Conversely, the NRT data accounted for 74.5% of its conflicts which occurred inside the boundaries of protected areas. This is likely due to the location and role of NRT, which is to monitor conservancies in Kenya's central region. The high level of conflicts in this area indicate that large amounts of conflicts are occurring inside conservancies, however they may not be recorded unless there is a specific group with the intentions of monitoring them. Finally, the UCDP data showed that 12.4% of all recorded conflicts occurred inside protected areas with 140 conflicts.

Overall, across the three datasets had 11.2% of the recorded conflicts occurred within the boundary of a protected area, indicating that 88.8% did not. While there were only 11.2% of the total conflicts which occurred inside protected areas, the stark difference in the number of recorded conflicts inside protected areas with the NRT data set compared to the other two reveals that there are conflicts occurring inside protected areas, however they may not be recorded due to selection bias and lack of media coverage in the collection process of conflict data for ACLED and UCDP.

After analyzing how many conflicts occurred inside protected areas, I also analyzed the count of each type of protected area that had conflicts recorded within their borders. Using the same designation modeled in the previous 2 figures, Figure 7 shows how many of each type of

protected area had recorded conflicts occur within its boundaries. Using the same methods that I used to calculate the information presented in table 1, I used QGIS modeling to analyze the types of protected areas with conflicts present which is shown in Figure 7. Similar to the other figures, not all types of protected areas are represented in the figure, only the ones with high conflict counts. In addition to the amount of each designation with one or more conflicts being modeled, I also added the number of each designation with zero recorded conflicts.

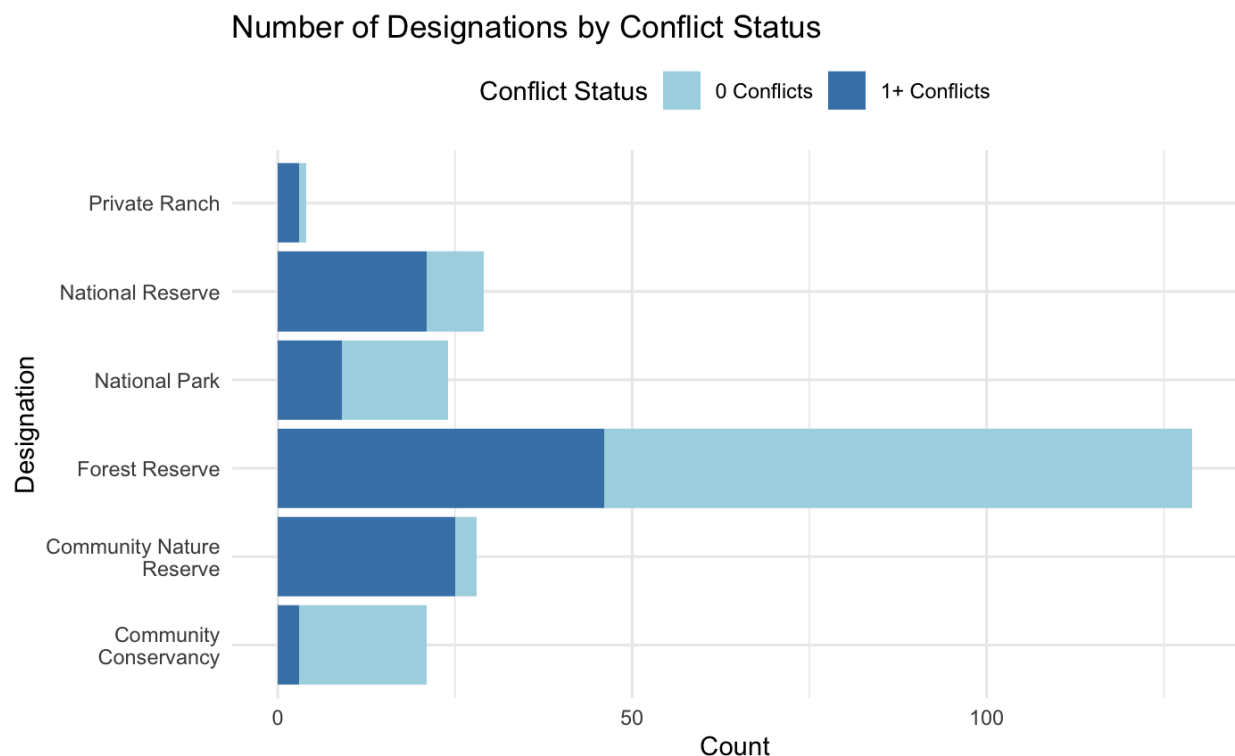


Figure 7 shows each designation of protected areas that had one or more conflicts occur within its boundaries and those that didn't. The dark blue represents the number of each designation with one or more conflicts recorded and the light blue represents the number of each designation with zero recorded conflicts.

Figure 7 shows the number of each designation of protected area where conflicts occurred and those where conflicts didn't occur. There were 44 forest reserves across Kenya which had at least one recorded conflict. National reserves and community nature reserves had over 20 of each which had conflicts present. On the other hand, community conservancies had a

relatively low number of those with recorded conflicts. When comparing Figure 7 with Figure 6, which shows how many of each type of protected area are present in Kenya, the majority of community nature reserves(89%) and national reserves(76%) had conflicts present within their boundaries. Additionally, around 19% of forest reserves had conflicts occur within their boundaries. The results from Figure 7 indicate that the main designations of protected areas which have a high number of conflicts are community nature reserves, forest reserves, and national reserves. These results follow my hypothesis, which was that community based protected areas, like community conservancies and community nature reserves, would have higher levels of recorded conflicts compared to other types of protected areas. Additionally, I hypothesized that reserves would have higher conflict levels, which is also present in Figure 7.

Another variable which is important to take into consideration when conducting a spatial analysis is area coverage and concentration of conflicts. While there are over 230 forest reserves in Kenya, many of these are smaller in size compared to other types of protected areas, like community nature reserves(Figure 5). Because of this, I calculated the number of conflicts per 100 square kilometers for each designation and mapped it to see those with the highest conflict levels. Figure 8 shows the average conflicts per 100 square kilometers for each designation with 95% confidence intervals. As a reference point, the designation labeled Kenya shows the average number of total conflicts across all 3 data sets for every 100 square kilometers..

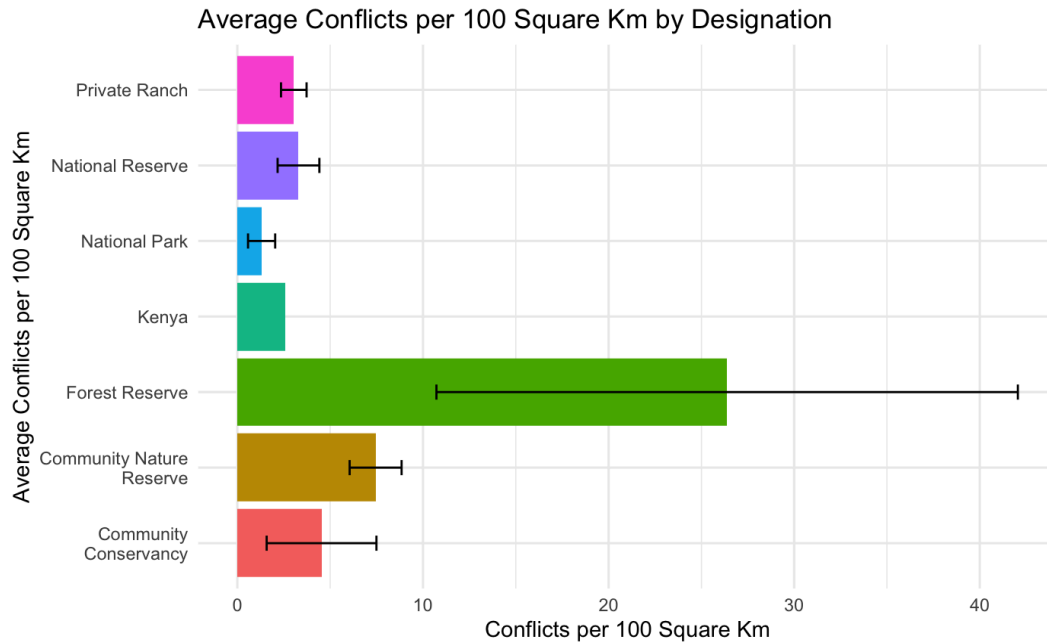


Figure 8 shows the average amount of conflicts per 100 square kilometers by designation. The error bars represent the 95% confidence interval.

The figure shows that forest reserves had the highest average conflicts per square kilometers, with around 26 conflicts/square km. However this value had a significantly large error bars, so it could range anywhere from 10 to 40 conflicts per kilometer. Community nature reserves had an average of 7.5 conflicts per square meter, community conservancies had an average of 4.5 conflicts per square kilometer, and national reserves had an average of 3.5 conflicts per square kilometer.

Looking at these averages compared to the number of conflicts per square kilometer across all of Kenya, community nature reserves and forest reserves were the only values which were significantly greater than the Kenyan average, because their error bars do not intersect with Kenya's average. The other designations with high concentration of conflicts are community conservancies, national reserves, and private ranches which have error bars that intersect with Kenya's average, but still have larger average conflicts per 100 square kilometers, indicating

high concentrations in these areas. Overall, the figure shows that community conservancies, community nature reserves, forest reserves, and national reserves are protected areas with the highest concentration of conflicts, indicating that conflict mitigation strategies should be focused on these designations.

This chapter highlights that protected areas definitely play a role in the climate change and conflict dynamic, due to the high amounts of recorded conflicts that occurred across the different designations, which were modeled in the last few figures. The results from this chapter follow along with my original hypotheses, which stated that community conservancies and reserves would have higher recorded conflicts compared to other types of protected areas. In the next chapter, I will look more into this dynamic, exploring how pastoralism is involved in inciting the conflicts that are occurring within protected areas.

Chapter 6: Pastoral Violence

While the last chapter explored the relationship between protected areas and conflict, there is evidence in previous research that this dynamic may be stimulated by pastoral violence. Therefore, this chapter will explore the similar data which has been analyzed in previous chapters, but in relation to pastoralism in order to gain a better understanding of what is driving pastoral violence in Kenya. I will be using the same data from the previous chapters, including the three conflict data sets and the protected area boundary data, in addition to a data set formulated by Nathan Nunn and Eoin McGuirk in their paper on transhumant pastoralism in Africa. In their data set, as stated in the methodology section, they created an 0.5 degree grid cell format across Africa and each cell had values assigned to it, like dependence on pastoralism as a livelihood, phytomass, and temperature. For this thesis, I will be using their dependence on pastoralism index and their phytomass index. I will be cross referencing this data with the protected area boundary data and the conflict data sets. This will help gain a better understanding of how pastoralism is intertwined in conflicts that are occurring within protected areas. I expect pastoralism to be heavily involved in protected area conflicts due to previous research which attributed many conflicts in Kenya to pastoralism, especially in the form of cattle raiding.

6.1 Protected Areas and Phytomass

To begin the chapter, I will analyze phytomass levels in protected areas compared to areas which are not considered protected areas across Kenya. Using the data from Nunn and McGuirk, which has phytomass values in a grid format, I will compare phytomass values inside protected areas compared to outside protected areas. This analysis is important because

pastoralists prefer to graze on land with higher phytomass values. If protected areas have higher phytomass values, then it indicates that pastoralists would have an incentive to graze on protected areas, which could be a mechanism for conflict. For the analysis, I used QGIS to find the grid cells which were inside the boundaries of protected areas and those that weren't and averaged the phytomass values. I ran a Welch two sample t-test on the dataset which is illustrated in Table 5.

Table 5: T-test Results for Phytomass in Protected vs. Non-Protected Areas

Metric	Value
T-value	-10.165
Degrees of Freedom (df)	473.16
P-value	< 2.2e-16
Confidence Interval	[-14.7855, -9.9952]
Mean in Group 0 (Outside)	28.38401
Mean in Group 1 (Inside)	40.77436

Table 5 shows the results from the t-test on the average phytomass calculated within the boundaries of protected areas and the average phytomass outside the boundaries of protected areas. The table also has relevant statistics including the p-value and confidence intervals.

The results from Table 5 show that there is a strong statistical significance between the average phytomass inside protected areas compared to the average phytomass outside protected areas. The average phytomass value outside protected areas is 28.38 while the average phytomass value inside protected areas is 40.77. Additionally, the confidence interval is from -14.7 to -9.99 which doesn't intersect with 0 and indicates statistical significance. Further proving this claim, the p-value is 2.2e-16 which is much less than 0.05 and shows a strong correlation between phytomass levels and protected areas. Additionally, this data was analyzed over the course of around 30 years, indicating that this trend has been present and robust throughout the past few decades. Overall, these results show a statistically significant difference

between phytomass inside protected areas and those outside. This correlation is important when analyzing the impact of protected areas on pastoral violence, which will be conducted in the next sections of this chapter, because pastoralists are incentivized to graze on land with high phytomass levels.

6.2 Protected Areas and Pastoralism

Next, I used QGIS to map the average pastoralism dependence index over Kenya in order to analyze the intersection between dependence of pastoralism and protected areas. Using the dataset from Nunn and McGuirk, I used their pastoralism index which represented how much an area, or grid cell, depends on pastoralism as a livelihood. Higher values are associated with higher dependency. I mapped the pastoralism and agricultural values on top of a map of Kenya in order to see whether areas which rely on pastoralism intersect or fall near protected areas. Figure 9 shows the results from this analysis, the red areas represent cells which have a higher dependency on pastoralism and the yellow polygons represent protected areas.

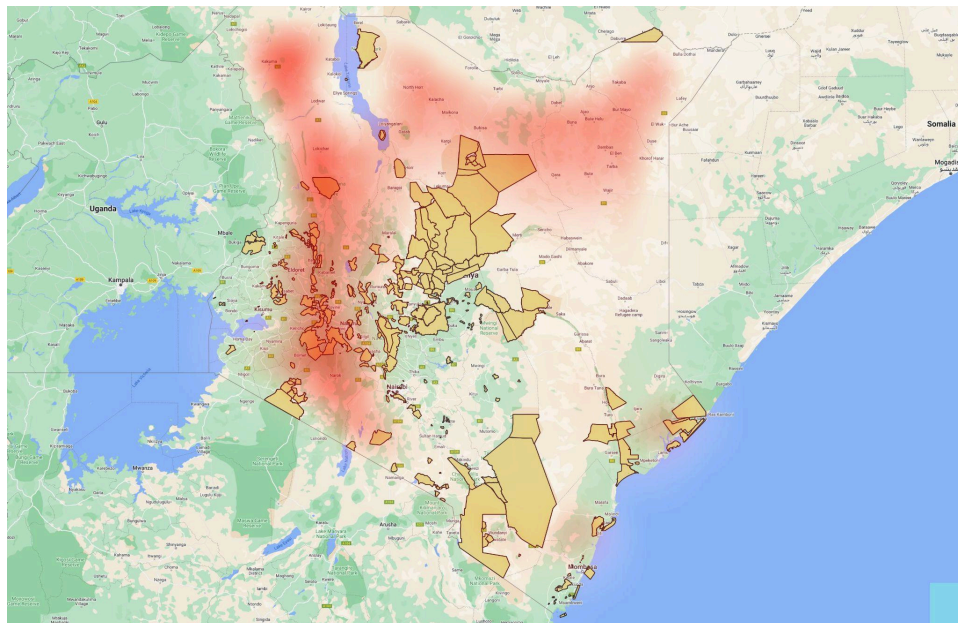


Figure 9 shows dependence on pastoralism across Kenya based on 0.5 degree grid cells. The yellow polygons represent protected areas.

Pastoralism dependence levels are mapped in Figure 9 on top of protected area boundary data to see whether protected areas intersect with higher pastoralism levels. As shown in the left side of Figure 9, which maps pastoralism values, the redder areas occur in the north and western parts of Kenya. Many of the smaller protected areas, most of which are forest reserves, in the west near Lake Victoria, intersect with areas that have high dependence on pastoralism. Some of the northern areas with higher dependence on pastoralism also intersect with the community nature reserves. Due to these results, I would hypothesize that the majority of pastoral conflicts that intersect with protected areas occur around the forest reserves in the western parts of Kenya. To test this hypothesis, in the next section I will isolate conflicts that are related to pastoralists from the ACLED data set and see how these conflicts intersect with protected areas.

6.3 Pastoral Violence

6.3.1 Pastoral Violence over Time

Using the ACLED data set, I isolated conflicts that are related to pastoral violence by filtering for keywords like “pastoralist” and “herder” in order to isolate conflicts which led to around 800 conflicts. The ACLED data set contains information on actors who are involved in each conflict, so I was able to refine the conflict data set to only include conflicts which had pastoralists or herders as one of the actors involved in the violence. I wasn’t able to replicate this method for the other data sets due to their limited information on the individual actors and details about each conflict. I understand that this method may leave out certain conflicts or include conflicts that aren’t necessarily considered pastoral violence, however it allows for a large-scale conflict analysis at the national level. Using the ACLED modified data set on pastoral violence, I mapped the conflicts over time to analyze general trends which is shown in Figure 10.

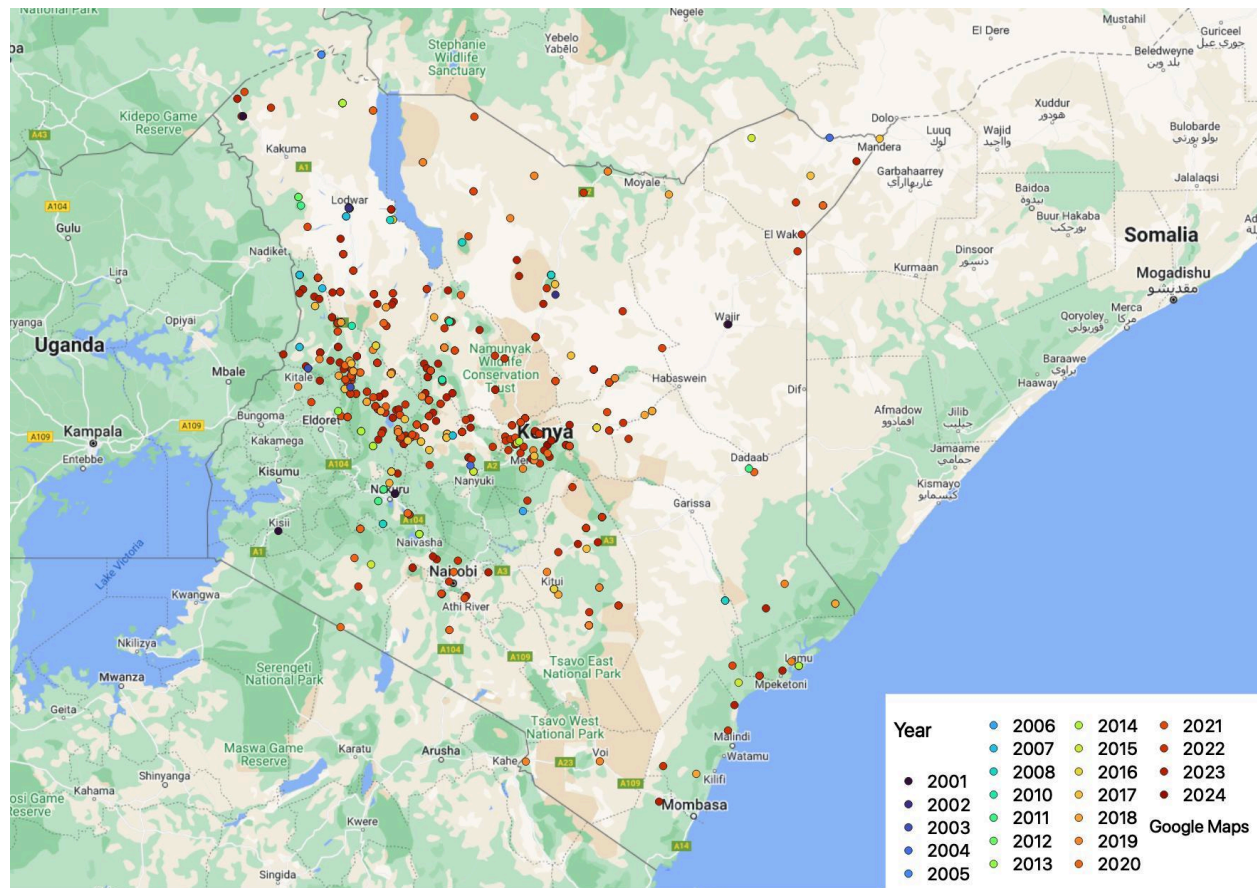


Figure 10 shows pastoral violence in Kenya by year. Each dot represents an individual conflict and the different colors correspond to different years.

Figure 10 shows how pastoral conflicts have evolved over the years and the spatial trends in these conflicts across Kenya. The figure shows that the majority of these conflicts occurred in recent years, with the majority of them occurring after 2017. Many of those which occurred recently are clustered in central and northern Kenya. On the other hand, the older conflicts are scattered around the rest of Kenya with no general trends.

Due to the stark number of pastoral conflicts which have occurred in recent years compared to earlier, the results indicate that pastoral violence is becoming a much more pressing issue. Due to this, I conducted an analysis on the percentage of total conflicts which involved

pastoralists each year and graphed it in Figure 11 to show how pastoral violence levels have evolved over time.

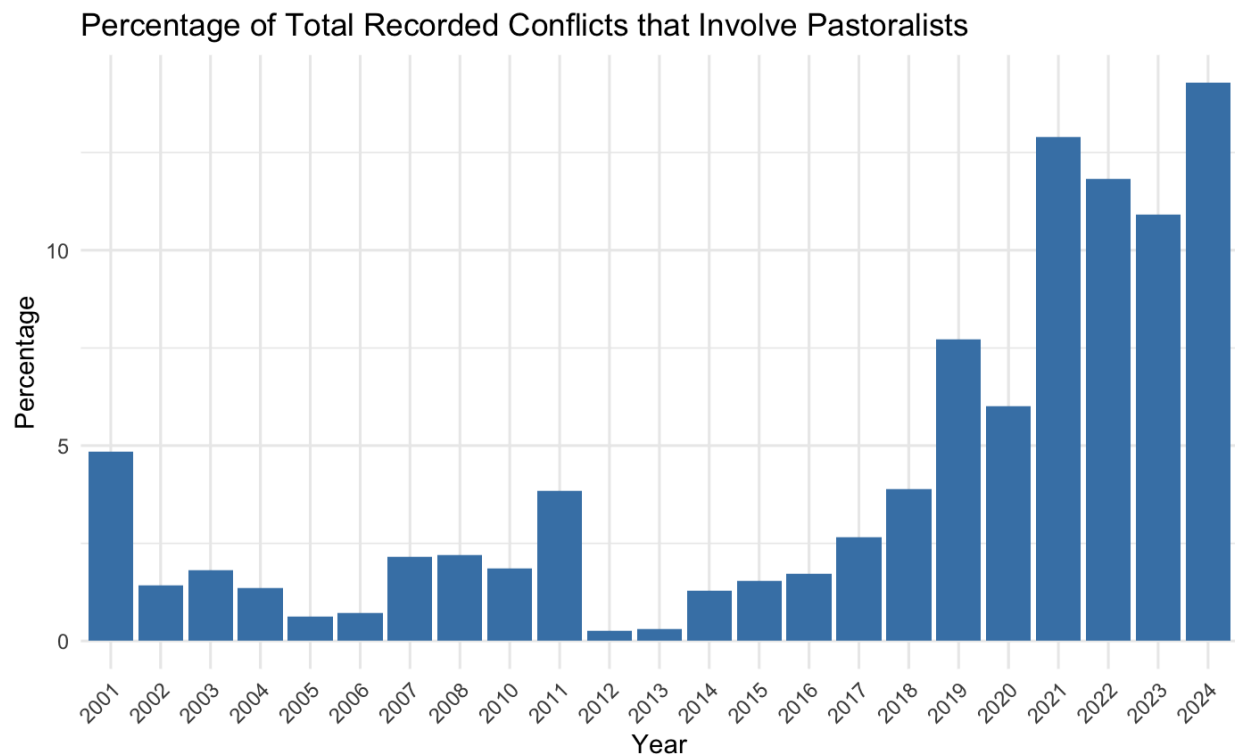


Figure 11 shows the percentage of conflicts in the ACLED data set which had pastoralists or herders as one of the two actors associated with the conflicts. Years not graphed had 0 conflicts.

Figure 11 illustrates a similar trend to that of Figure 10 indicating that pastoral violence has increased in recent years, especially since 2018 when the percentage of conflicts that involved pastoralists started significantly increasing. While it may seem like a small percentage of the total conflicts recorded by ACLED, the increasing trend shows that the issue will most likely only get more severe in the coming years. This indicates the need for conflict mitigation strategies targeting pastoral violence.

6.3.2 Pastoral Violence and Rainfall

Given the trends explored in Chapter 4, which found a correlation between higher rainfall and conflicts by year, I analyzed this same trend with pastoral violence. I ran a linear regression,

similar to that of Chapter 4, using annual rainfall in mm/year as the independent variable and pastoral conflicts by year as the dependent variable. The results of this regression are shown in Table 6.

Table 6: Regression Results Summary of Pastoral Violence and Annual Rainfall

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	744.8554	34.0193	21.90	<0.0001
Pastoral Conflicts	1.5042	0.6422	2.34	0.0302*

Table 6 shows the results from the linear regression between annual rainfall in mm/year and pastoral conflicts per year.

The linear regression shown in Table 6 shows a significant correlation between higher annual rainfall in mm/year and pastoral conflicts in the ACLED data set. For a one deviation increase in annual rainfall, one mm/year, there is a 1.5 unit increase in pastoral conflicts per year. This positive correlation follows a similar trend to that of Chapter 4, but has a stronger correlation between rainfall and violence, suggesting that more pastoral violence occurs in years with higher rainfall. As I explained in Chapter 4, I analyzed the regression annually rather than monthly in order to analyze general trends and more long term effects of higher rainfall. These results align with my original hypothesis, which suggests that pastoral violence will occur more frequently in years with higher rainfall, due to livestock raiding.

6.3.3 Pastoral Violence within Protected Areas

In addition to the previous section, I also analyzed pastoral violence inside protected areas. I conducted a similar analysis to that of Chapter 5, where I used QGIS modeling to calculate the number of individual conflicts which occurred within protected areas and which type of designation of protected areas had recorded conflicts. Overall, I found that 106 of the 769 pastoral conflicts occurred within protected areas, which is 13.7% of the total conflicts. In the

previous chapter, I found that only 6.4% of the total ACLED conflicts occurred within protected areas. This indicates that there is a much higher percentage of pastoral conflicts occurring within protected areas compared to all types of conflicts. Looking at the type of protected areas which had pastoral violence occurring within their boundaries, I graphed the amount of each designation of protected areas which had one or more conflicts occur within their boundaries which is shown in Figure 12. The designations which are graphed all had one or more pastoral conflicts occur within their boundaries. The designations not graphed had no pastoral conflicts recorded.

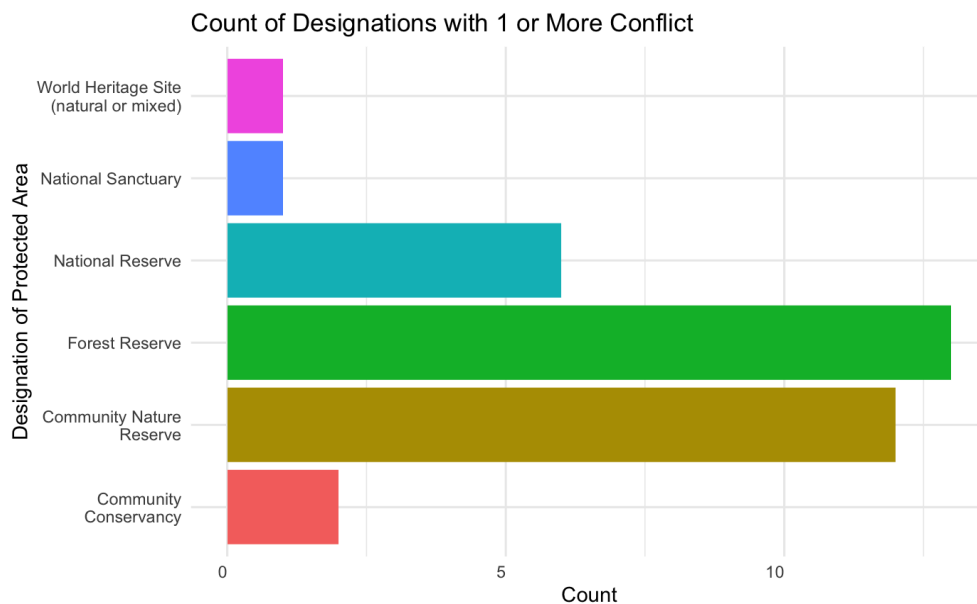


Figure 12 shows the count of each designation of protected area which had 1 or more pastoral conflicts occur within its boundaries. Designations not graphed had no recorded pastoral conflicts.

As shown in Figure 12, forest reserves, community nature reserves, and national reserves had the highest count of recorded conflicts. These results are consistent with previous analyses, which found forest reserves, community nature reserves, and national reserves to have high levels of conflict, with 5 to 12 of each designation having at least one recorded conflict.

Community conservancies, national sanctuaries, and world heritage sites had a few recorded conflicts as well.

Overall, the results from this chapter follow along with my original hypotheses. To begin, protected areas were found to have higher phytomass values than areas which are not protected. Additionally, pastoral violence has been increasing over time and occurs in years with higher rainfall, which has been attributed to the opportunistic mindset of livestock raiders. Finally, community based protected areas and reserves had the highest count of recorded pastoral conflicts. In the next chapter, I will discuss the results from the previous analyses.

Chapter 7: Discussion

7.1 Key Findings

Climate change and conflict research has yet to come to a consensus on whether climate change drives conflict, and the present research further adds to this inconsistency, linking higher rainfall and lower drought levels to more conflict. The results from Chapter 4 show that for annual rainfall and drought levels, lower visible impacts of climate change on the environment are linked with higher conflict levels. This relationship was present across two of the three data sets, and overall had fairly low p values, indicating a significant relationship. While this correlation was not what I hypothesized, there are a few possible explanations for why higher rainfall and lower drought was found to correlate with more conflicts.

One explanation for this relationship, which has been researched in Northern Kenya, is that during years with higher rainfall there is more livestock raiding due to the opportunistic mindset of cattle raiders.¹³⁴ When there is higher rainfall, there is more available pasture, and more pastoralists taking advantage of the pasture, which offers cattle raiders the opportunity to steal livestock, which increases violent conflict.¹³⁵ In addition, these raids often have political or ethnic motivations associated with them, rather than being triggered by competition over scarce resources.¹³⁶ When I isolated conflicts which contained pastoralists as one of the main actors involved, the correlation between higher rainfall and conflict was even more significant than the

¹³⁴ Detges, "Close-up on Renewable Resources and Armed Conflict."

¹³⁵ Witsenburg and Adano, "Of Rain and Raids."

¹³⁶ "Climate Clashes? Weather Variability, Land Pressure, and Organized Violence in Kenya, 1989–2004 - Ole Magnus Theisen, 2012," accessed December 3, 2023, <https://journals.sagepub.com/doi/10.1177/0022343311425842>.

previous models, suggesting that this relationship is attributed to cattle raiding. Oftentimes, cattle raiding is used as a political weapon and involves heavy calculations and planning.¹³⁷

Another explanation which could explain the positive correlation between rainfall and conflict, could be that climate change may not be directly impacting conflict levels, but climate variation may be. As explained in the literature review, climate variation, which means greater extremes of climate data, has been correlated with increased conflict. Kenya has experienced high levels of climate variation in recent years, which could be a mechanism contributing to conflict. Climate variation leaves Kenya with longer periods of more intense rain and longer periods of drought, which has negatively affected pastoralists because it leaves them with longer periods of limited fresh pasture, which negatively impacts the health of their herds.¹³⁸ The average annual rainfall and drought levels accounted for climate change rather than climate variation, so one direction for future research could be to analyze the impact of climate variation on conflict.

The analysis which I conducted with Nunn and McGuirk's grid cell data on rainfall and phytomass also suggests that livestock raiding is a major factor contributing to the positive relationship between rainfall and conflict. The maps which I created from the rainfall and phytomass levels indicated that many of the recorded conflicts intersected with areas of higher rainfall and higher phytomass. While the results from these models can't directly prove causation between livestock raiding and higher rainfall, the presence of previous research and strong correlations suggests that this relationship is significant.

Introducing protected areas as a variable in my analysis revealed their significant role in the complex interplay of climate change and conflict. Protected areas don't have fencing or any

¹³⁷ Greiner, "Guns, Land, and Votes."

¹³⁸ "Absorbing Climate Shocks and Easing Conflict in Kenya's Rift Valley | Crisis Group."

physical boundaries, which allows pastoralists, wildlife, and anyone else to use protected area land with little monitoring.¹³⁹ In addition, reserves allow for limited human consumption and usage of the reserves' natural resources. Due to this, I anticipated reserves and community conservancies to have higher recorded conflicts compared to national parks and other types of protected areas. In my analysis this proved to be true and forest reserves, national reserves, and community nature reserves had the highest levels of recorded conflicts within their boundaries. While there have been many efforts attempting to build cohesion and engage communities in conservation efforts, sometimes it has led to more conflicts or hasn't had an effect on decreasing conflicts.¹⁴⁰ This suggests that there is a need for a new set of policies directed towards reducing conflicts in reserves and community based protected areas like conservancies.

When I analyzed this relationship with respect to conflicts per 100 square kilometers, I found a similar trend with community based protected areas and reserves having more concentrated conflicts. I found that community conservancies, community nature reserves, national reserves, and forest reserves had higher conflicts per 100 square kilometers compared to the average conflicts per 100 square kilometers in Kenya. This further proves that there is a need for more solutions directed towards mitigating conflicts in these protected areas. Additionally, there is a lack of research looking at the specific dynamics of conflicts between community conservancies and reserves at the individual level in Kenya, so it is hard to create solutions geared towards mitigating these conflicts when there is limited information on how these conflicts are arising.

¹³⁹ "Overview | Kenya Wildlife Service."

¹⁴⁰ Clemens Greiner, "Unexpected Consequences: Wildlife Conservation and Territorial Conflict in Northern Kenya," *Human Ecology* 40, no. 3 (June 1, 2012): 415–25, <https://doi.org/10.1007/s10745-012-9491-6>.

In the final results chapter, I found a statistically significant difference between the average phytomass values inside protected areas and outside protected areas. This outcome was anticipated, considering the fundamental purpose of protected areas: the conservation of nature and its resources.¹⁴¹ However, the stark difference between phytomass levels in protected areas and the phytomass levels outside protected areas further incentivizes pastoralists to graze on protected area land. Due to the lack of physical barriers, which are intended to allow wildlife to roam where they please, pastoralists can graze on protected areas with little disturbance. Additionally, I found that a significant portion of land in Kenya which is heavily dependent on pastoralism as a main livelihood intersected with many of the forest reserves in western Kenya. Similarly, forest reserves had the highest count of recorded pastoral conflicts. This suggests that pastoralism is heavily intertwined in protected area conflicts, especially in forest reserves.

Using the new data set which isolated pastoral violence, I found that pastoral violence has been significantly increasing over time. The percentage of ACLED conflicts which included pastoralists as one of the main actors involved has increased from 2001 to 2024. Additionally, many of these conflicts which have been recorded recently occurred in central and western portions of Kenya, which are areas with high concentrations of protected areas. This indicates that pastoral conflicts are recently becoming concentrated in the central western portion of Kenya near many of the protected areas. This hypothesis follows the explanation which I explained recently, which attributes many recent conflicts to livestock raiding.¹⁴²

These findings suggest a potential area for further investigation: the role of protected areas, with their abundant phytomass, in attracting pastoralists to graze on these lands,

¹⁴¹“Effective Protected Areas | IUCN,” accessed April 1, 2024, <https://www.iucn.org/our-work/topic/effective-protected-areas>.

¹⁴²Witsenburg and Adano, “Of Rain and Raids.”

particularly during the wetter months when they contain a surplus of fresh pasture. This scenario, in turn, appears to provide an opportunity for cattle raiders, who exploit the increased livestock presence to intensify their attacks.

The final portion of my results section analyzed pastoral conflicts which were occurring within protected areas and found that forest reserves, national reserves, and community nature reserves had the highest count of recorded conflicts. This follows similar patterns to the previous analysis. The percentage of total conflicts which occurred inside protected areas was around 6.4%, while the percentage of pastoral conflicts which occurred inside protected areas was 13.7%, which is a significantly higher proportion.

To summarize, these results suggest that pastoralism is intertwined in conflicts across Kenya, and is a potential reason for the positive correlation between rainfall and conflict. This thesis adds to the climate conflict literature proving that the issue of conflict in Africa is one that can't simply be attributed to climate change. However, it also adds to the research on pastoral violence, which attributed this positive correlation to livestock raiding. This thesis suggests a similar pattern to that of cattle raiding, due to the high significance value of pastoral violence and rainfall regression. Furthermore, forest reserves, community nature reserves, and community conservancies had high levels of conflicts, for all types of conflicts and pastoral conflicts, suggesting that these types of protected areas should be targeted when looking at conflict mitigation approaches in protected areas. Overall, these results suggest that pastoral violence and conflicts in protected areas are still occurring today at high rates, so the need for strategies geared towards mitigating these conflicts is paramount. In the next section of the discussion, I will outline limitations of this thesis and potential directions for future research.

7.2 Limitations

The majority of studies which explore climate change and conflict in Africa focus on the regional level or multi-national level¹⁴³, however a limited amount of research focuses on exploring this dynamic within individual countries. Due to this, I chose to focus on the national scope for this study, looking at conflicts across Kenya rather than all of Africa or a specific area within Kenya. This scope helped gain a better understanding of large scale trends in conflicts in Kenya and it gave a large enough sample size so that it could be subsetting for conflicts inside protected areas and conflicts which involved pastoralists.

While looking at conflicts at the national level assisted with my understanding of conflicts in Kenya, one major limitation of the study is that its results are less applicable to other countries. Kenya's political landscape, high density of protected areas, and high dependence on pastoralism as a major livelihood create a unique mix of dynamics which give this study a low external validity, meaning it is less likely to be applied to other countries. On the other hand, a smaller scope would have helped understand individual dynamics and specific types of conflict, like the livestock raiding which is occurring in Northern Kenya.

Another main limitation of the study is the coverage of the datasets which were being used. The three major datasets which I used were inconsistent in their spatial coverage and the amount of data associated with each conflict. In the future, using one data set which has a more equal coverage across Kenya and a uniform identification system associated with each conflict would allow for more robust results. The NRT data set, which had the majority of its conflicts occur inside protected areas, suggested that conflicts are occurring inside protected areas,

¹⁴³“Research Methods for Exploring the Links between Climate Change and Conflict - Ide - 2017 - WIREs Climate Change - Wiley Online Library.”

however they were underreported by the ACLED and UCDP data sets. Using a more uniform data set which emphasizes pastoral violence and protected areas would help with gaining a better understanding of pastoral violence.

7.3 Recommendations for Future Research

Using the results from this study, which explored conflicts at a national level with respect to protected areas and pastoral violence, there are many additional pathways to be explored which can give a more comprehensive analysis of specific conflicts in Kenya. For example, conducting an ethnographic study to understand the specificities of cattle raiding and whether cattle raiders do tend to take advantage of pastoralists during wetter months could help understand how to mitigate these conflicts in the future. Furthermore, exploring similar variables to this study, like conflict levels, but with respect to climate variation rather than climate change could be another direction for future research. While I used rainfall and drought levels, looking at climate variation levels by year could also yield interesting linear regression results. Additionally, analyzing this relationship with time periods less than a year could help understand short term trends in conflicts. Another potential direction for future research could be an ethnographic or dataset study specifically focusing on conflicts occurring inside protected areas in Kenya.

Furthermore, the results from this thesis showed that forest reserves, national reserves, and community nature reserves had the highest levels of conflicts, so exploring conflicts in these specific areas could also help gain a better understanding of what solutions can be done to mitigate these conflicts. In the next chapter, I will outline a set of policy recommendations

geared towards reducing conflict across Kenya, specifically with respect to protected areas and pastoral violence.

Chapter 8: Policy Recommendations

In this chapter, I will present a series of policy recommendations derived from the findings of this thesis. These suggestions are tailored for a range of stakeholders, including the Kenyan government and the international community, local communities and governments, protected area management bodies, and pastoralist communities. In formulating these recommendations, I have carefully considered their practicality and the lessons learned from previously tested solutions to ensure that my proposed strategies are viable. To begin, I will first review the conflict mitigation strategies previously implemented in Kenya.

Silvestri et. al (2012) analyzed key adaptation strategies of pastoralist communities against climate change through diversifying livestock feeds, destocking, and migrating livestock to other sites. However, they found that these adaptation strategies faced additional challenges of livestock markets and lack of land access.¹⁴⁴ Additionally, scholars have found that climate change adaptation strategies have been effective in increasing food security of pastoralists. Early warning systems and communication to pastoralists of weather information were two ways in which pastoralists increased their food security.¹⁴⁵

One of the main issues with implementing climate change and conflict mitigation strategies in pastoralist communities is the lack of agency that pastoralists are given in the policy making process. Songok et. al (2011) highlights the need for agro-pastoralists to have a say in the decision making process in order to create proper climate change adaptation and disaster

¹⁴⁴Silvia Silvestri et al., “Climate Change Perception and Adaptation of Agro-Pastoral Communities in Kenya,” *Regional Environmental Change* 12, no. 4 (December 1, 2012): 791–802, <https://doi.org/10.1007/s10113-012-0293-6>.

¹⁴⁵S. Wagura Ndiritu and Geoffrey Muricho, “Impact of Climate Change Adaptation on Food Security: Evidence from Semi-Arid Lands, Kenya,” *Climatic Change* 167, no. 1 (July 28, 2021): 24, <https://doi.org/10.1007/s10584-021-03180-3>.

reduction strategies.¹⁴⁶ Furthermore, indigenous communities have been found to have a surplus of knowledge on climate change adaptation strategies, suggesting the importance of including them in the decision making process.¹⁴⁷

For protected areas, conservancies have previously attempted to integrate pastoralists into the tourism industry as an addition to their pastoral income. Researchers found that tourism provided pastoralists with a stable, year round income, however the conservancies restricted livestock mobility and access to grazing land.¹⁴⁸ Consequently, the integration of conservancies with pastoralist communities, though promising in theory, led to significant trade-offs, ultimately diminishing the economic productivity of pastoralism. Nevertheless, these previously implemented solutions lay a solid groundwork for my recommendations. By making minor adjustments and introducing new recommendations, stakeholders can focus on developing infrastructure, embracing indigenous knowledge, and effectively distributing information to pastoral communities.

8.1 Recommendations for the Kenyan Government and the International Community

8.1.1 Humanitarian aid to pastoralists

Considering the detrimental effects of climate variability and drought on pastoralist livelihoods, it's critical to provide economic assistance to pastoralists during periods of heightened climate

¹⁴⁶ Charles Kipkorir Songok et al., “Improving the Participation of Agro-Pastoralists in Climate Change Adaptation and Disaster Risk Reduction Policy Formulation: A Case Study from Keiyo District, Kenya,” in *Experiences of Climate Change Adaptation in Africa*, ed. Walter Leal Filho (Berlin, Heidelberg: Springer, 2011), 55–68, https://doi.org/10.1007/978-3-642-22315-0_4.

¹⁴⁷ Agricultural Sector Development Support Programme (ASDSP), Samburu County, Kenya et al., “An Evaluation of Climate Change Indigenous Coping and Adaptation Strategies for Sustainable Agro-Pastoral Based Livelihoods in Baringo County, Kenya,” *IOSR Journal of Environmental Science, Toxicology and Food Technology* 8, no. 8 (2014): 38–58, <https://doi.org/10.9790/2402-08833858>.

¹⁴⁸ Bedelian and Ogutu, “Trade-Offs for Climate-Resilient Pastoral Livelihoods in Wildlife Conservancies in the Mara Ecosystem, Kenya.”

variation and drought. This assistance could take the form of cash transfers or the implementation of more sustainable water management practices. Typically, during dry seasons, a significant portion of water resources is allocated to farmers and communities, often leaving pastoralists and their livestock with insufficient water. Providing economic aid and enhancing water management strategies are essential for ensuring that pastoralists can effectively manage water resources and receive adequate support throughout the dry season, a time when their economic productivity is notably impacted by the adverse consequences of climate change. Furthermore, for pastoralists facing financial setbacks due to wildlife in protected areas, offering humanitarian assistance could foster improved relations between pastoralists and the management of protected areas.

8.1.2 Invest in Climate Adaptation Measures

The government and international community are urged to back the adoption of climate adaptation strategies throughout Kenya, with a focus on regions susceptible to drought and scant rainfall. Possible interventions include the introduction of drip irrigation systems for farmers, the establishment of water storage facilities in anticipation of the wet season, and the promotion of drought-resistant grass varieties. Such measures would bolster the resilience of pastoralists and farmers during dry periods when their means of subsistence are under threat. Additionally, these strategies could contribute to conflict reduction by diminishing the competition for scarce resources.

8.1.3 Assign Land Specifically for Pastoralism

A potential approach the Kenyan government and water management authorities could pursue involves designating fenced areas exclusively for pastoralism. The root of many conflicts in Kenya today can be traced back to ambiguous land ownership; therefore, marking specific

territories for pastoral use could alleviate such disputes. This strategy would also curtail the grazing of livestock on lands designated as protected areas. To ensure the effectiveness of these designated lands, adequate funding and maintenance are essential, necessitating both international and government support. Moreover, it's imperative that these areas have independent water sources to maintain fresh pastures for grazing.

8.1.4 Youth Employment and Limited Arms Dealing

Efforts to increase youth employment across Kenya, supported by both international and governmental initiatives, could serve as a potential deterrent to cattle raiding, a significant issue highlighted in this thesis. Such interventions could provide young men, who are typically involved in these raids, with alternative sources of income, thereby reducing their involvement in seizing livestock herds. Furthermore, restricting young individuals' access to firearms could also diminish the frequency of cattle raiding, as cattle raiders often rely on guns during these incidents, leaving pastoralists defenseless and compelled to surrender their livestock.

8.1.5 Creation and Dispersal of Climate Change and Conflict Data

This thesis underscores the crucial role of data in shedding light on the underlying factors of conflicts, their locations, and the parties involved, thereby aiding in devising and applying conflict resolution strategies. The establishment of a comprehensive dataset on conflicts and climate change, supported by the international community and the Kenyan government, could significantly enhance the ability of pastoralists, farmers, communities, and researchers to pinpoint the causes of conflicts. Additionally, insights from climate change data and modeling can inform pastoralists and farmers about the timing and intensity of droughts, enabling better preparation. Research into conflict dynamics can guide communities and local governments in formulating effective mitigation approaches. Moreover, sharing weather forecasts and

establishing early warning systems can provide pastoral communities with vital information for drought readiness and climate adaptation efforts.

8.2 Recommendations for Communities and Local Governments

8.2.1 Community Run Initiatives to De-Escalate Tensions

In the event of conflicts, there's often an absence of measures aimed at calming tensions, and when official entities do intervene, their attempts may be received with skepticism or resistance. Hence, it's beneficial for communities themselves to spearhead efforts designed to mitigate conflicts and soothe tensions. Given the diversity in tribal and community norms across Kenya, tailoring these initiatives to align with the unique customs and practices of each community is crucial. Additionally, indigenous communities possess extensive knowledge about climate change and pastoralism, which should be integrated into the policymaking process to ensure decisions are informed and culturally sensitive.

8.2.2 Infrastructure Building in Pastoralist Heavy Communities

Local governments and communities can play a significant role in reducing conflicts by enhancing infrastructure to better support pastoralists during intense droughts. This support could encompass educational initiatives, improved water management systems, healthcare services, and financial assistance. In many instances, assistance that comes directly from the community is more favorably accepted than that from government or official bodies, particularly in the context of Kenya's volatile political environment. Community-led efforts to develop infrastructure and provide aid to pastoralists are likely to be more warmly welcomed and effective compared to initiatives led solely by local governmental authorities.

8.3 Recommendations for Protected Area Management

8.3.1 Participatory Tourism Model

The presence of conflicts within protected areas highlights the need for a stronger relationship between these areas and the surrounding communities. A participatory tourism model could serve as a means to enhance this relationship, enabling both communities and pastoralists to contribute to the management decisions of protected areas. While some participatory tourism models have already been enacted in conservancies in Kenya, these models fail to include community members in the decision making process. Considering the significant land use tensions in Kenya, empowering communities through active participation in decision-making can potentially reduce conflicts and foster a sense of support. Moreover, community members possess extensive knowledge about local wildlife, which could enrich the tourism model of protected areas by incorporating traditional practices and cultural heritage.

8.3.2 Education and Employment Opportunities

Enhancing support for communities and pastoralists near protected areas can also be achieved by providing educational and employment opportunities. Creating job openings for pastoralists, especially during the dry season, presents an alternative means of economic support during critical times. Furthermore, educating the communities about the significance and benefits of protected areas can help ease the tensions between these areas and the surrounding populations. Such initiatives promote a sense of collaboration and foster a reciprocal relationship, benefiting both protected area management and the local communities.

8.4 Recommendations for Pastoralists

8.4.1 Creation of Migratory Pastoralist Corridors

Considering the scarcity of fresh grazing lands for pastoralists, who are frequently compelled to venture into protected areas with high levels of phytomass, establishing migratory corridors offers a viable solution for fostering a symbiotic relationship. These corridors would consist of designated strips of land spanning both protected and private territories, specifically set aside for pastoral use. Local authorities or community groups could oversee the creation and management of these passageways, aiming to reduce conflicts and enable pastoralists to access lands within protected areas without significantly disrupting wildlife habitats.

Chapter 9: Conclusion

“In northern Kenya, we come from many different communities that all practice pastoralism and that all face similar challenges, but we will not be able to solve these issues — of climate change, the loss of our grazing lands and having a future for our people — unless we can solve our conflicts peacefully and be united.”

Ramson Karmushu¹⁴⁹

While the results from this thesis may seem like pastoral violence and protected area conflicts are only a small piece of the big puzzle that is conflict in Africa, however the research also shows that pastoral violence and the effects of climate change will only continue to increase. As humans, animals, and the environment become more disconnected as entities, climate change continues to increase, and humans continue to cause conflict, the issues outlined in this thesis are only going to become more severe. The discussion section highlights just a few of the many problems that need to be addressed in Kenya regarding violence, but with proper attention and support to Kenya, these problems can be addressed one at a time.

The results of this thesis show the nuanced relationship between climate change and conflict, suggesting that there are an additional surplus of factors driving conflicts in Kenya in addition to climate change. Two of these which are outlined in this thesis are dependence on pastoralism as a livelihood and protected areas. Both of these require careful consideration when implementing conflict mitigation strategies across Kenya. Protected areas, despite being originally introduced to increase conservation efforts, further escalated land use tensions. On the other hand, pastoralism is a vital part of Africa's cultural and ecological diversity. As we work to address the impacts of climate change on pastoral communities, we must recognize the value of

¹⁴⁹ “Kenya: As Drought Deepens Land Conflicts, Peacebuilders Respond,” United States Institute of Peace, accessed April 1, 2024, <https://www.usip.org/blog/2022/09/kenya-drought-deepens-land-conflicts-peacebuilders-respond>.

their traditional practices and the importance of integrating their knowledge into conservation and development efforts.

Appendix

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This thesis represents my own work in accordance with University Regulations.

/s/ Meera Burghardt